

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 1 — Product Philosophy & Operating Principles

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1. Vision

Day Zero OS is **not** a task manager.

It is **not** a Notion clone.

It is **not** a Jira clone.

It is **not** a second brain.

It is an **Operating System for Builders**.

A builder should be able to manage an entire startup, product, YouTube channel, research, learning journey, and personal execution without leaving Day Zero OS.

Every feature must answer one question:

Does this help a builder execute better?

If not, it does not belong.

2. Design Philosophy

The interface should feel like:

- Linear
- Raycast
- Notion
- GitHub
- VS Code

combined together.

Not visually.

Functionally.

The application should prioritize:

- Speed
 - Clarity
 - Focus
 - Minimal friction
 - Deep organization
-

3. Product Principles

Every module follows these principles.

Principle 1

Everything belongs to a Project.

Examples:

Projects

↓

Knowledge

↓

Assets

↓

Content

↓

Architecture Decisions

↓

Activity

↓

Milestones

Nothing exists in isolation.

Projects are the center of the OS.

Principle 2

Every action creates history.

Example:

Create Project

↓

Activity Timeline

Edit Description

↓

Activity Timeline

Upload Asset

↓

Activity Timeline

Publish Content

↓

Activity Timeline

Delete Milestone

↓

Activity Timeline

Nothing important happens silently.

Principle 3

Everything is connected.

Knowledge references Projects.

Projects reference Assets.

Assets reference Content.

Content references Knowledge.

Architecture Decisions reference Development.

No duplicated information.

Only relationships.

Principle 4

Fast Capture.

The user should never spend more than a few seconds creating something.

Example:

New Project

↓

Enter Name

↓

Create

↓

Continue working

Everything else can be edited later.

Principle 5

Progressive Detail.

Users first create.

Then enrich.

Not the opposite.

Never force long forms.

Principle 6

Everything is editable.

No locked pages.

No permanent decisions.

Every field.

Every section.

Every card.

Every item.

Should be editable.

Principle 7

Delete is always available.

Every entity supports:

Create

Read

Update

Delete

unless there is a strong reason not to.

Deleting should require confirmation.

Principle 8

Archive instead of destroy.

Whenever possible:

Archive

instead of

Delete.

Users can restore archived items later.

Principle 9

No dead screens.

There should never be:

Coming Soon

Placeholder

Under Construction

Empty White Page

Instead:

Show helpful empty states.

Example:

"No milestones yet."

↓

"Create your first milestone."

with a button.

Principle 10

Every page has purpose.

Opening any page should immediately answer:

Why does this page exist?

What can I do here?

What should I do next?

4. UX Rules

Every page must contain:

Loading State

Empty State

Error State

Success State

Confirmation Dialog

Keyboard Navigation

Responsive Layout

5. CRUD Rule

Every data entity supports:

Create

Read

Update

Delete

Search

Filter

Sort

Bulk Operations (where applicable)

Undo (where practical)

6. Navigation Rules

Maximum three clicks to reach anything.

Primary Navigation:

Mission Control

Projects

Content

Knowledge

Assets

Weekly Debrief

Settings

Search (Ctrl/Cmd + K)

7. Data Ownership

Every user owns:

Workspace

↓

Projects

↓

Everything else

No orphan records.

No floating data.

8. AI Philosophy

AI is optional.

AI is disabled by default.

The OS must remain fully usable without AI.

AI should enhance workflows, not replace them.

No feature should require AI.

9. Quality Standards

Every feature must:

Persist to Supabase

Support edit

Support delete

Handle failures gracefully

Provide user feedback

Maintain activity history

Pass build

Pass lint

Pass typecheck

10. Definition of Complete

A feature is **NOT** complete when the UI appears.

A feature is complete only when:

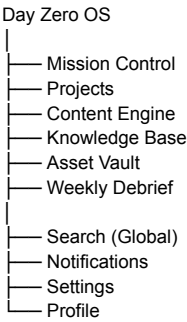
- UI is implemented
- CRUD is implemented
- Validation exists
- Empty state exists
- Loading state exists
- Error state exists
- Activity logging exists
- Navigation works
- Supabase persistence works
- Build passes
- Lint passes
- Typecheck passes

Only then is it considered complete.

Chapter 2 — Complete Application Architecture & Navigation Map

2.1 Application Hierarchy

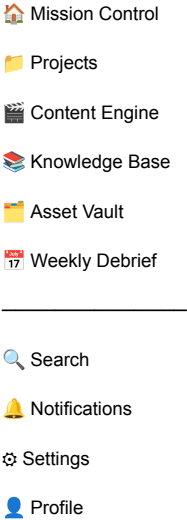
Day Zero OS is organized into six primary domains.



Everything in the application belongs to one of these domains.

2.2 Global Navigation

The left sidebar is the permanent navigation.



The sidebar must remain consistent throughout the application.

2.3 Mission Control

Purpose:

The homepage of Day Zero OS.

It answers one question:

What should I work on right now?

Mission Control contains:

Today's Focus

Current Project

Current Video

Today's Tasks

Upcoming Deadlines

Recent Activity

Quick Actions

Project Health

Content Pipeline

Weekly Progress

Recent Projects

Recent Knowledge

Recent Assets

Recent Decisions

Mission Control never stores data.

It only summarizes information from the rest of the OS.

2.4 Projects

Projects are the core entity.

Every other module connects to a Project.

Projects

└─ Project List

└─ Create Project

└─ Archive

└─ Trash

└─ Templates

└─ Favorites

Each Project opens the Project Workspace.

2.5 Project Workspace

Every project contains:

Project Workspace

└─ Overview

└─ Planning

└─ Development

└─ Knowledge

└─ Assets

└─ Content

└─ Architecture

└─ Activity

└─ Settings

This is the heart of Day Zero OS.

2.6 Overview

Purpose:

Project homepage.

Contains:

Project Information

Description

Priority

Status

Progress

Technology

Deadline

Team (future)

Cover Image

Tags

Recent Activity

Quick Links

Quick Statistics

2.7 Planning

Purpose:

Plan execution.

Contains:

Milestones

Tasks

Sprint Goals

Roadmap

Calendar

Dependencies

Timeline

2.8 Development

Purpose:

Engineering workspace.

Contains:

Development Notes

GitHub Repository

Repository URL

Branch

Commit Reference

Bug List

Technical Debt

Sprint Notes

Deployment Notes

Testing Checklist

2.9 Knowledge

Purpose:

Project-specific second brain.

Contains:

Markdown Notes

Research

Reference Links

Meeting Notes

Prompt Library

Bookmarks

Learning

Ideas

2.10 Assets

Purpose:

Central file manager.

Contains:

Images

Videos

PDFs

Documents

Brand Assets

Icons

Fonts

Logos

Figma Files

GitHub Links

External URLs

2.11 Content

Purpose:

Manage all content related to the project.

Pipeline:

Idea

↓

Research

↓

Outline

↓

Script

↓

Recording

↓

Editing

↓

Thumbnail

↓

SEO

↓

Publishing

↓

Analytics

2.12 Architecture

Purpose:

Record important technical decisions.

Each entry contains:

Decision

Reason

Alternatives

Impact

Related Files

Date

Author

These act like permanent engineering records.

2.13 Activity

Purpose:

Chronological history of the project.

Automatically records:

Project Created

Milestone Added

Milestone Completed

Knowledge Added

Asset Uploaded

Decision Created

Content Published

Project Updated

Nothing is entered manually.

2.14 Content Engine (Global)

Purpose:

View all content across all projects.

Views:

Board

Calendar

Table

Published

Draft

Analytics

2.15 Knowledge Base (Global)

Purpose:

View all knowledge across projects.

Views:

All Notes

Research

Prompt Library

Bookmarks

Recently Edited

Favorites

2.16 Asset Vault (Global)

Purpose:

Central media library.

Views:

Grid

List

Recent Uploads

Images

Videos

Documents

Logos

Unused Assets

2.17 Weekly Debrief

Purpose:

Weekly reflection.

Contains:

Wins

Mistakes

Lessons

Productivity

AI Discoveries

Hours Worked

2.18 Search

Global search.

Keyboard shortcut:

Ctrl + K

or

⌘ + K

Searches:

Projects

Knowledge

Assets

Content

Architecture Decisions

Commands

Settings

2.19 Notifications

Contains:

Project Updates

Due Milestones

Content Reminders

System Notifications

Success Messages

Warnings

2.20 Settings

Contains:

Profile

Workspace

Appearance

Theme

Notifications

Keyboard Shortcuts

AI

Storage

Export

Import

Danger Zone

2.21 Relationships

Workspace

↓

Projects

└─ Planning

└─ Development

└─ Knowledge

└─ Assets

└─ Content

└─ Architecture

└─ Activity

Global modules aggregate data from all projects.

Projects
↓
Content Engine

Projects
↓
Knowledge Base

Projects
↓
Asset Vault

Projects
↓
Mission Control

2.22 Navigation Rules

Every page must be reachable within **three clicks**.

Every entity links back to its parent.

No dead ends.

Every page includes:

- Breadcrumb navigation
- Back button
- Search access
- Current project context (where applicable)

Chapter 3 — Mission Control (Complete Functional Specification)

3.1 Purpose

Mission Control is the **homepage of Day Zero OS**.

It should never be a dashboard that simply displays data.

It should answer one question immediately:

"What should I do next?"

The screen should eliminate decision fatigue.

When the application opens, the user must instantly understand:

- Current priorities
- Active project
- Deadlines
- Progress
- Work requiring attention
- Recent work
- Suggested next action

Mission Control never creates data.

It only summarizes and visualizes data from all other modules.

3.2 Screen Layout

Mission Control contains the following sections.

Mission Control

Today's Mission

Quick Actions

Current Project

Current Video

Upcoming Deadline

Weekly Progress

Project Health

Content Pipeline

Today's Tasks
Recent Activity
Recent Decisions
Recent Projects
Recent Knowledge
Recent Assets
System Status

3.3 Today's Mission

Purpose
Displays today's primary objective.
Examples
Finish Project Workspace
Record Episode 3
Complete Sprint Planning
Publish YouTube Video
Functions
- Edit - Save - Clear - Auto-update from Weekly Debrief (optional)
Database
user_settings.today_mission

3.4 Quick Actions

Purpose
Fast access to frequently used actions.
Buttons
New Project
New Knowledge
Upload Asset
New Content
New Decision
New Milestone
Requirements
Every button opens its creation modal immediately.
Keyboard shortcuts
Ctrl + Shift + P
New Project
Ctrl + Shift + N
New Note
Ctrl + Shift + U
Upload
Ctrl + Shift + C
New Content

3.5 Current Project Card

Shows

- Project Name
- Status
- Progress
- Priority
- Deadline
- Last Edited

Buttons

Open

Edit

Archive

Delete

Clicking the card opens Project Workspace.

3.6 Current Video

Displays

Current YouTube episode.

Fields

Title

Stage

Estimated Publish Date

Progress

Thumbnail

Buttons

Open

Edit

Continue Writing

3.7 Upcoming Deadlines

Shows

Next five deadlines.

Fields

Title

Project

Due Date

Days Remaining

Priority

Sorting

Ascending.

Color Rules

Overdue

Red

3 Days

Orange

7 Days

Yellow

Safe

Green

3.8 Weekly Progress

Shows

Projects Completed

Videos Published

Knowledge Added

Assets Uploaded

Milestones Completed

Hours Worked

Visualization

Progress bars.

3.9 Project Health

Calculated automatically.

Inputs

Missed Deadlines

Completed Milestones

Progress %

Activity Frequency

Open Tasks

Outputs

Excellent

Good

Needs Attention

Critical

3.10 Content Pipeline

Shows

Ideas

Research

Outline

Script

Recording

Editing

Published

Each stage displays count.

Click opens Content Engine.

3.11 Today's Tasks

Displays

Tasks due today.

Supports

Complete

Edit

Delete

Reschedule

Completion automatically updates activity.

3.12 Recent Activity

Displays latest actions.

Examples

Project Created

Milestone Completed

Knowledge Added

Asset Uploaded

Video Published

Decision Added

Click navigates to related item.

3.13 Recent Decisions

Displays

Latest Architecture Decisions.

Fields

Decision

Project

Date

Impact

Click opens Decision.

3.14 Recent Projects

Displays

Last five opened projects.

Functions

Open

Pin

Archive

Delete

3.15 Recent Knowledge

Displays

Recently edited notes.

Click opens Markdown editor.

3.16 Recent Assets

Displays

Recent uploads.

Supports

Preview.

Download.

Open.

Delete.

3.17 System Status

Displays

Supabase

Connected

Authentication

Online

Storage

Usage

AI

Disabled / Enabled

Last Sync

Purpose

Instant system diagnostics.

3.18 CRUD

Mission Control itself creates nothing.

Everything is read-only except:

Today's Mission.

Quick Actions.

3.19 Empty States

Examples

No active projects.

Create your first project.

[Create Project]
No content yet.

Start your first video.

[Create Content]

3.20 Loading

Use skeleton cards.

Never blank screen.

3.21 Error State

Show

Unable to load projects.

Retry

Never show raw errors.

3.22 Database Sources

Mission Control reads from

projects

milestones

content_items

activity_logs

architecture_decisions

knowledge_entries

assets

weekly_debriefs

user_settings

3.23 Completion Criteria

Mission Control is complete when

- ✓ Every widget loads real Supabase data
- ✓ Every Quick Action works
- ✓ Clicking every card navigates correctly
- ✓ All statistics calculate automatically
- ✓ Empty states exist
- ✓ Loading states exist
- ✓ Error states exist
- ✓ Build passes
- ✓ Lint passes
- ✓ Typecheck passes

END OF CHAPTER 3

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 4 — Projects Module (Complete Functional Specification)

4.1 Purpose

Projects are the **central entity** of Day Zero OS.

Everything belongs to a Project.

Knowledge.

Assets.

Content.

Architecture.

Milestones.

Activity.

Nothing exists independently.

4.2 Projects Screen Layout

Projects

Search

Filters

Sort

Views

Project List

Pagination

Bulk Actions

4.3 Header Actions

Buttons

New Project

Import

Templates

Archived

Trash

4.4 Project Card

Displays

Cover Image

Project Name

Status

Priority

Progress

Deadline

Technology

Recent Activity

Actions

Open

Edit

Duplicate

Archive

Delete

Favorite

4.5 Create Project

Minimum fields

Project Name

Optional

Description

Priority

Status

Technology

Tags

Deadline

Cover

Quick create should remain the default.

4.6 Edit Project

Editable

Everything.

Autosave supported.

4.7 Delete Project

Confirmation Dialog

Delete Project?

This action cannot be undone.

Cancel

Delete

Delete removes

Knowledge

Assets

Milestones

Content

Activity

Architecture

through database relationships.

4.8 Archive

Moves project to Archived.

Restore available.

4.9 Duplicate

Creates full copy.

Option

Duplicate with Assets

Duplicate without Assets

4.10 Favorite

Pinned to Mission Control.

4.11 Search

Searches

Project Name

Description

Technology

Tags

Instant.

4.12 Filters

Status

Priority

Technology

Deadline

Archived

Favorites

Multiple filters allowed.

4.13 Sorting

Recently Updated

Recently Created

Name

Deadline

Priority

Progress

Ascending/Descending.

4.14 Views

Grid

List

Board

Timeline

4.15 Bulk Actions

Supports

Archive

Delete

Export

Favorite

Tag

4.16 Empty State

No Projects

Create your first project.

[New Project]

4.17 Loading

Skeleton project cards.

4.18 Error State

Unable to load projects.

Retry

4.19 Database Behaviour

Reads

projects

Writes

projects

activity_logs

Updates

Mission Control automatically.

4.20 Navigation

Click Project

↓

Open Project Workspace.

4.21 Completion Criteria

Projects module is complete when

- ✓ Create
- ✓ Read
- ✓ Update
- ✓ Delete
- ✓ Archive
- ✓ Duplicate
- ✓ Search
- ✓ Filters
- ✓ Sort
- ✓ Bulk Actions
- ✓ Views
- ✓ Favorites
- ✓ Empty State
- ✓ Loading State
- ✓ Error State
- ✓ Activity Logging
- ✓ Supabase Persistence
- ✓ Build
- ✓ Lint
- ✓ Typecheck

Chapter 5 — Project Workspace (Complete Functional Specification)

5.1 Purpose

The Project Workspace is the **heart of Day Zero OS**.

Everything related to a project is managed here.

The Project Workspace should become the place where a builder spends **90% of their time**.

The user should never need to leave the workspace to manage a project.

5.2 Navigation

Project Workspace

Overview

Planning

Development

Knowledge

Assets

Content

Architecture

Activity

Settings

Top Navigation:

← Back to Projects

Project Name

Favorite

Archive

Delete

Search Inside Project

Share (Future)

5.3 Workspace Header

Contains

Project Cover

Project Name

Status

Priority

Progress

Technology

Deadline

Created Date

Updated Date

Quick Actions

Edit

Archive

Delete

Duplicate

Favorite

Export

5.4 Workspace Sidebar

Persistent.

Displays

Overview

Planning

Development

Knowledge

Assets

Content

Architecture

Activity

Settings

Shows badge counts.

Example

Knowledge (18)

Assets (42)

Content (6)

Milestones (12)

5.5 Project Context

Every page inside Workspace knows

Project ID

Workspace ID

Project Name

Current User

Never ask user to select project again.

5.6 Universal Project Toolbar

Appears on every tab.

Contains

Search

Quick Add

Notifications

Recent Activity

Command Palette

Project Settings

5.7 Quick Add Menu

Available everywhere.

+

↓

Milestone

Task

Knowledge

Asset

Content

Decision

Note

5.8 Project Search

Search only inside current project.

Searches

Knowledge

Assets

Content

Architecture

Milestones

Activity

5.9 Project Statistics

Visible in Workspace.

Completion %

Milestones

Knowledge

Assets

Videos

Recent Activity

Health

Open Issues

5.10 Workspace State

Every tab supports

Loading

Empty

Success

Error

5.11 CRUD Rule

Every section supports

Create

Read

Update

Delete

Unless explicitly read-only.

5.12 Navigation Rule

Switching tabs never loses data.

Unsaved changes prompt

Leave?

Save

Discard

Cancel

5.13 Autosave

Supported for

Description

Notes

Markdown

Research

Development Notes

Scripts

Every 5 seconds after changes.

5.14 Activity Logging

Every important action logs automatically.

Examples

Description Updated

Milestone Added

Knowledge Created

Asset Deleted

Decision Edited

Video Published

5.15 Database Relationships

Workspace reads

projects

milestones

knowledge_entries

assets

content_items

architecture_decisions

activity_logs

5.16 Keyboard Shortcuts

- Ctrl+S
 - Save
 - Ctrl+K
 - Search
 - Ctrl+/ - Command Palette
 - Ctrl+Shift+N
 - New Note
 - Ctrl+Shift+M
 - New Milestone
-

5.17 Completion Criteria

Workspace is complete when

- ✓ Navigation works
 - ✓ Every tab loads
 - ✓ Shared toolbar works
 - ✓ Search works
 - ✓ Quick Add works
 - ✓ Activity logging works
 - ✓ Autosave works
 - ✓ Supabase integration works
-

END OF CHAPTER 5

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 6 — Overview Module (Complete Functional Specification)

6.1 Purpose

Overview is the **homepage of a Project**.

It answers

What is the current state of this project?

A builder should understand the project within **10 seconds**.

6.2 Layout

Overview

Cover Image

Project Information

Progress

Health

Deadline

Description

Technology

Tags

Quick Links

Recent Activity

Statistics

6.3 Cover Image

Functions

Upload

Replace

Remove

Preview

Supported

PNG

JPG

WEBP

6.4 Project Information

Fields

Project Name

Description

Status

Priority

Category

Technology

Tags

Start Date

Deadline

Created

Updated

Every field editable.

6.5 Progress

Displays

0–100%

Slider

Numeric

Progress Bar

Automatic calculation optional.

Manual override allowed.

6.6 Project Health

Calculated from

Progress

Missed Deadlines

Activity

Milestones

Open Bugs

States

Excellent

Good

Warning

Critical

6.7 Description

Markdown supported.

Functions

Edit

Autosave

Fullscreen

Preview

6.8 Technology Stack

Supports

React

Next.js

Supabase

Python

Docker

Node

Custom

Searchable tags.

6.9 Tags

Unlimited.

Colors supported.

Examples

AI

Startup

YouTube

MVP

Learning

6.10 Quick Links

Supports

GitHub

Figma

Website

Documentation

Drive

Notion

External URLs

Functions

Add

Edit

Delete

Open

6.11 Statistics

Displays

Milestones

Knowledge Notes

Assets

Videos

Architecture Decisions

Activity Count

Click opens related section.

6.12 Recent Activity

Last 10 activities.

Open

Filter

View All

6.13 Favorite

Pin project.

Mission Control reads favorites.

6.14 Export

Supports

JSON

Markdown

PDF

ZIP (Future)

6.15 Validation

Required

Project Name

Optional

Everything else.

6.16 Loading State

Skeleton cards.

6.17 Empty State

No Description

Add a project description.

[Edit]

No Links

Add your first repository.

[Add Link]

6.18 Error State

Unable to load project.

Retry

6.19 CRUD

Supports

Edit

Save

Delete Cover

Replace Cover

Add Tag

Delete Tag

Add Link

Delete Link

Update Progress

6.20 Database

Reads

projects

Writes

projects

activity_logs

6.21 Completion Criteria

Overview is complete when

- ✓ All fields editable
 - ✓ Autosave works
 - ✓ Cover uploads
 - ✓ Links work
 - ✓ Statistics update
 - ✓ Activity logs generated
 - ✓ Supabase persistence
 - ✓ Loading
 - ✓ Empty
 - ✓ Error
 - ✓ Build
 - ✓ Lint
 - ✓ Typecheck
-

Chapter 9 — Knowledge Module (Complete Functional Specification)

9.1 Purpose

The Knowledge Module is the **second brain** of every project.

It stores everything that helps build the project:

- Research
- Ideas
- Documentation
- Tutorials
- API References
- Meeting Notes
- AI Prompts
- Learnings
- Decisions
- Bookmarks

Nothing should be lost.

The user should never think:

"Where did I save that note?"

9.2 Layout

Knowledge

Search

Categories

Tags

Views

Knowledge List

Editor

Linked Resources

Related Notes

9.3 Views

Support:

Grid

List

Tree

Timeline

Favorites

9.4 Categories

Every note belongs to a category.

Default categories:

Research

Ideas

Documentation

Meeting Notes

Tutorials

API Docs

AI Prompts

References

Learning

Bookmarks

Personal Notes

Users can create custom categories.

9.5 Knowledge Note

Each note contains:

Title

Category

Markdown Content

Tags

Created Date

Updated Date

Favorite

Pinned

Project

Author

Attachments

Related Notes

Related Assets

9.6 Markdown Editor

Supports:

Headings

Lists

Tables

Code Blocks

Images

Links

Task Lists

Block Quotes

Horizontal Rules

Callouts

Emoji

Mermaid Diagrams (Future)

Functions:

Edit

Preview

Split View

Fullscreen

Word Count

Character Count

Reading Time

Autosave

9.7 Attachments

Attach:

Images

PDF

Videos

ZIP

Documents

External URLs

Actions:

Preview

Open

Download

Replace

Delete

9.8 AI Prompt Library

A special category.

Each prompt contains:

Title

Prompt

Purpose

Tags

Model Used

Last Used

Favorite

Functions:

Copy

Edit

Duplicate

Delete

Pin

9.9 Bookmarks

Store:

Websites

YouTube

GitHub

Papers

Blogs

Courses

Display:

Title

URL

Description

Category

Thumbnail

Added Date

9.10 Search

Searches:

Title

Markdown

Tags

Category

Attachments

Prompt Content

Instant search.

9.11 Filters

Category

Tags

- Favorites
 - Pinned
 - Recently Edited
 - Recently Created
-

9.12 Relationships

Every note can link to:

- Project
- Assets
- Content
- Architecture Decisions
- Other Notes

Backlinks should be displayed automatically.

9.13 CRUD

Supports:

- Create
 - Read
 - Update
 - Delete
 - Duplicate
 - Archive
 - Favorite
 - Pin
 - Export
-

9.14 Export

Formats:

- Markdown
 - PDF
 - JSON
-

9.15 Empty State

No Knowledge Yet

Capture your first idea.

[New Note]

9.16 Loading

Skeleton list.

9.17 Error

Unable to load knowledge.

Retry

9.18 Database

Reads:

knowledge_entries

Revision Required:

knowledge_categories
knowledge_tags
knowledge_links
bookmarks
prompt_library

9.19 Activity

Automatically records:

Knowledge Created

Knowledge Updated

Knowledge Deleted

Bookmark Added

Prompt Saved

9.20 Completion Criteria

Knowledge Module is complete when:

- Markdown works
- Autosave works
- Search works
- Categories work
- Tags work
- Attachments work
- Prompt Library works
- Bookmarks work
- CRUD complete
- Export complete
- Activity logging complete
- Supabase persistence complete

END OF CHAPTER 9

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Chapter 10 — Asset Vault (Complete Functional Specification)

10.1 Purpose

The Asset Vault is the **Digital Asset Management (DAM)** system for Day Zero OS.

It stores every file related to every project.

The user should never need to search their Downloads folder again.

10.2 Layout

Asset Vault

Search

Filters

Views

Folders

Assets Grid/List

Preview Panel

Storage Statistics

10.3 Supported Asset Types

Images

Videos

Audio

PDF

Word

Excel

PowerPoint

ZIP

Code Files

Fonts

Icons

Logos

SVG

JSON

Markdown

Design Files

External Links

10.4 Asset Card

Displays:

Preview

File Name

File Type

Size

Project

Upload Date

Tags

Storage Location

10.5 Upload

Supports:

Click Upload

Drag & Drop

Paste Image

Import URL

10.6 Preview

Images

Videos

PDF

Markdown

Text Files

Audio

Preview opens inside the application.

No download required.

10.7 File Actions

Open

Preview

Download

Rename

Replace

Duplicate

Move

Copy Link

Delete

Archive

10.8 Folder Management

Supports:

Create Folder

Rename

Move

Delete

Nested Folders

Color Labels

10.9 Tags

Unlimited tags.

Color coded.

Searchable.

10.10 Search

Searches:

File Name

Tags

Project

Extension

Folder

Description

10.11 Filters

File Type

Project

Upload Date

Recently Opened

Favorites

Unused Assets

Large Files

10.12 Views

Grid

List

Gallery

Timeline

10.13 Storage

Displays:

Total Storage

Used Storage

Remaining Storage

Largest Files

Recent Uploads

10.14 External Links

Supports:

GitHub

Figma

Google Drive

YouTube

Website

API Documentation

10.15 Image Viewer

Supports:

Zoom

Rotate

Fullscreen

Next

Previous

Download

10.16 CRUD

Supports:

Upload

Edit

Rename

Replace

Delete

Archive

Restore

10.17 Empty State

No Assets Yet

Upload your first file.

[Upload]

10.18 Loading

Upload progress.

Skeleton previews.

10.19 Error

Upload Failed

Retry

Cancel

10.20 Database

Current:

assets

Revision Required:

asset_folders
asset_tags
asset_links
asset_versions

10.21 Activity

Automatically logs:

Asset Uploaded

Asset Deleted

Asset Renamed

Folder Created

Folder Deleted

Asset Downloaded

10.22 Completion Criteria

Asset Vault is complete when:

- Upload works
 - Preview works
 - Image viewer works
 - Folder management works
 - Search works
 - Filters work
 - Tags work
 - Storage statistics work
 - CRUD complete
 - Activity logging complete
 - Supabase Storage integration complete
-

END OF CHAPTER 10

Database Revision Log

Revision 2

New tables introduced:

knowledge_categories
knowledge_tags
knowledge_links
bookmarks
prompt_library

asset_folders
asset_tags
asset_links
asset_versions

These will be included in the **final consolidated migration** after the PCS is complete.

Chapter 11 — Content Engine Module (Complete Functional Specification)

11.1 Purpose

The Content Engine manages the complete lifecycle of every piece of content created by the user.

This module is not limited to YouTube.

It should support future expansion to:

- YouTube
- LinkedIn
- X (Twitter)
- Instagram
- Blog
- Newsletter
- Podcast
- Documentation

For Version 1, YouTube is the primary workflow.

11.2 Layout

Content

Pipeline

Content List

Editor

Publishing

Analytics

11.3 Content Pipeline

Each content item moves through fixed stages.

Idea

↓

Research

↓

Outline

↓

Script

↓

Recording

↓

Editing

↓

Thumbnail

↓

SEO

↓

Ready

↓

Published

↓

Archived

Users can drag and drop between stages.

11.4 Content Item

Each content item contains

Title

Platform

Series

Episode Number

Project

Status

Priority

Publish Date

Description

Thumbnail

Tags

Keywords

Notes

11.5 Research

Contains

Research Notes

References

Links

Competitors

Videos

Articles

Papers

Screenshots

11.6 Outline

Supports

Sections

Bullet Points

Time Estimates

References

Objectives

11.7 Script Editor

Markdown editor.

Supports

Formatting

Code Blocks

Images

Tables

Callouts

Comments

Autosave

Version History

Statistics

Characters

Words

Estimated Duration

Reading Time

11.8 Recording

Track

Recording Date

Equipment

Camera

Microphone

Lighting

Location

Duration

Take Count

11.9 Editing

Fields

Editor

Software

Progress

Export Version

Notes

Revision History

11.10 Thumbnail

Stores

Thumbnail Image

PSD/Figma Link

Title Variations

Design Notes

11.11 SEO

Contains

Title

Description

Keywords

Tags

Hashtags

Chapters

Pinned Comment

11.12 Publishing

Stores

Platform

Publish Date

URL

Visibility

Playlist

Category

11.13 Analytics

Displays

Views

Watch Time

CTR

Subscribers Gained

Revenue (Future)

Comments

Likes

Manual entry for MVP.

11.14 Search

Searches

Title

Script

Tags

Platform

Series

11.15 Filters

Status

Platform

Series

Publish Date

Priority

11.16 CRUD

Supports

Create

Edit

Delete

Duplicate

Archive

Export

Move Between Stages

11.17 Activity

Automatically logs

Content Created

Script Updated

Recording Completed

Published

Thumbnail Updated

11.18 Empty State

No Content Yet

Create your first video.

[New Content]

11.19 Loading

Skeleton cards.

11.20 Error

Unable to load content.

Retry

11.21 Database

Current

content_items

Revision Required

content_series

content_versions

seo_metadata

content_analytics

11.22 Completion Criteria

Content Engine complete when

- ✓ Pipeline
- ✓ Script Editor
- ✓ Research
- ✓ SEO
- ✓ Publishing
- ✓ Analytics
- ✓ Search
- ✓ Filters
- ✓ CRUD
- ✓ Activity
- ✓ Supabase
- ✓ Build
- ✓ Lint
- ✓ Typecheck

END OF CHAPTER 11

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 12 — Architecture Decisions Module (Complete Functional Specification)

12.1 Purpose

This is one of the signature features of Day Zero OS.

Instead of forgetting why technical decisions were made, every important architectural decision is permanently recorded.

Think of it as a project's engineering memory.

12.2 Layout

Architecture Decisions
Decision Timeline
Decision List
Decision Details
Related Resources

12.3 Decision Record

Every decision contains

Title
Decision
Problem Statement
Reason
Alternatives Considered
Chosen Solution
Consequences
Impact
Status
Priority
Author
Created Date
Updated Date

12.4 Status

Proposed
Approved
Implemented
Deprecated
Rejected

12.5 Priority

Critical
High
Medium
Low

12.6 Related Items

Every decision may link to

Project
Milestone
Task
Knowledge Note
Asset
Content
GitHub Repository
Bug
Technical Debt

12.7 Decision Timeline

- Displays
- Newest
- Oldest
- Grouped by Sprint
- Grouped by Milestone

12.8 Templates

- Provide templates for
- Architecture
- Database
- UI
- Backend
- API
- Security
- Infrastructure
- Performance
- AI

12.9 Search

- Searches
- Title
- Decision
- Reason
- Alternatives
- Impact

12.10 Filters

- Status
- Priority
- Author
- Project
- Date

12.11 CRUD

- Supports
- Create
- Edit
- Delete
- Duplicate
- Archive
- Export

12.12 Export

- Formats
- Markdown
- PDF
- JSON

12.13 Activity

Automatically logs

Decision Created

Decision Updated

Decision Approved

Decision Archived

Decision Deleted

12.14 Empty State

No Architecture Decisions

Document your first important decision.

[New Decision]

12.15 Loading

Skeleton cards.

12.16 Error

Unable to load architecture decisions.

Retry

12.17 Database

Current

architecture_decisions

Revision Required

decision_templates

decision_links

12.18 Completion Criteria

- Architecture module complete when
- ✓ CRUD
 - ✓ Timeline
 - ✓ Templates
 - ✓ Search
 - ✓ Filters
 - ✓ Export
 - ✓ Related Resources
 - ✓ Activity
 - ✓ Supabase
 - ✓ Build
 - ✓ Lint
 - ✓ Typecheck

Chapter 13 — Activity Timeline Module (Complete Functional Specification)

13.1 Purpose

The Activity Timeline is the permanent history of a project.

It automatically records every important action performed inside Day Zero OS.

The user should never manually create an activity entry.

The timeline becomes the project's audit log and historical record.

13.2 Objectives

The Activity Timeline should answer:

- What happened?
- Who performed it?
- When did it happen?
- Why did it happen?
- Which project did it affect?
- Which module generated it?

13.3 Layout

Activity Timeline

Search

Filters

Timeline

Activity Details

Statistics

13.4 Timeline Card

Each activity contains

Icon

Title

Description

Module

Project

Timestamp

User

Related Resource

Example

 Project Created

Day Zero OS

2 minutes ago

Open

13.5 Activity Types

Support

Project

Milestone

Task

Knowledge

Content

Asset

Decision

Bug

Deployment

Settings

Authentication

AI

Import

Export

13.6 Generated Activities

Examples

Project Created

Project Updated

Project Deleted

Milestone Added

Milestone Completed

Task Completed

Knowledge Created

Knowledge Edited

Asset Uploaded

Asset Deleted

Decision Approved

Content Published

Bug Closed

Deployment Updated

Profile Updated

13.7 Activity Details

Opening an activity displays

Summary

Previous Value

New Value

Related Module

Open Resource

Timestamp

User

13.8 Search

Searches

Title

Description

Project

Module

User

13.9 Filters

Today

Yesterday

Last Week

Project

Module

User

Success

13.10 Grouping

Support

Today

Yesterday

This Week

This Month

Older

13.11 Timeline Views

Chronological

Grouped

Compact

Detailed

13.12 Statistics

Displays

Today's Activities

Weekly Activities

Most Active Project

Most Active Module

Average Daily Activity

13.13 Navigation

Every activity links back to its origin.

Example

Activity

↓

Project

↓

Knowledge

↓

Specific Note

13.14 Export

Formats

JSON

CSV

PDF

Markdown

13.15 CRUD

Activities are immutable.

Supports

Read

Search

Filter

Export

No Edit.

No Delete.

(Except system cleanup.)

13.16 Empty State

No Activity Yet

Start building.

Everything you do will appear here.

13.17 Loading

Skeleton timeline.

13.18 Error

Unable to load activity.

Retry

13.19 Database

Current

activity_logs

Revision

activity_metadata

13.20 Completion Criteria

Activity Timeline complete when

- ✓ Automatic logging
 - ✓ Search
 - ✓ Filters
 - ✓ Timeline
 - ✓ Navigation
 - ✓ Export
 - ✓ Statistics
 - ✓ Supabase
 - ✓ Build
 - ✓ Lint
 - ✓ Typecheck
-

END OF CHAPTER 13

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 14 — Global Content Engine (Cross-Project Content Management)

14.1 Purpose

Unlike the Project Workspace Content tab, this module manages **all content across all projects**.

It provides a centralized production dashboard.

14.2 Layout

Content Engine

Pipeline

Calendar

Content Library

Analytics

Publishing Queue

14.3 Dashboard

Displays

Total Videos

Drafts

Recording

Editing

Ready

Published

Scheduled

14.4 Pipeline Board

Kanban

Ideas

↓

Research

↓

Outline

↓

Script

↓

Recording

↓

Editing

↓

Thumbnail

↓

SEO

↓

Ready

↓

Published

Drag & Drop supported.

14.5 Calendar View

Shows

Recording

Editing

Publishing

Deadlines

Views

Daily

Weekly

Monthly

14.6 Content Library

Each card displays

Thumbnail

Title

Project

Series

Stage

Publish Date

Platform

14.7 Publishing Queue

Displays

Ready

Scheduled

Failed

Published

Supports manual publish status for MVP.

14.8 Analytics Dashboard

Shows

Views

Subscribers

CTR

Watch Time

Comments

Likes

Manual entry for Version 1.

Future API integrations.

14.9 Search

Searches

Title

Project

Series

Tags

Platform

14.10 Filters

Project

Stage

Platform

Series

Published

Scheduled

14.11 Views

Board

Table

Calendar

Gallery

14.12 Bulk Actions

Archive

Delete

Move Stage

Export

Duplicate

14.13 Templates

Templates for

Tutorial

Course

Vlog

Podcast

Product Demo

Shorts

Announcement

14.14 Relationships

Every content item links to

Project

Knowledge

Assets

Architecture Decisions

Scripts

14.15 Empty State

No Content Available

Create your first content item.

[Create]

14.16 Loading

Skeleton board.

14.17 Error

Unable to load content engine.

Retry

14.18 Database

Current

content_items

content_series

content_analytics

Revision

content_templates

publishing_queue

14.19 Completion Criteria

Global Content Engine complete when

✓ Dashboard

✓ Pipeline

✓ Calendar

✓ Analytics

✓ Search

✓ Filters

✓ Templates

✓ Publishing Queue

✓ CRUD

✓ Activity

✓ Supabase

✓ Build

✓ Lint

✓ Typecheck

Chapter 15 — Global Knowledge Base (Cross-Project Knowledge Management)

15.1 Purpose

The Global Knowledge Base is the **master knowledge hub** of Day Zero OS.

Unlike the Project Workspace Knowledge tab, this module provides a unified view of knowledge across **all projects**.

It should become the user's permanent knowledge repository.

The user should never need to remember which project contains a specific note.

15.2 Objectives

The Knowledge Base should answer:

- What do I already know?
 - Where is that information?
 - Which project uses it?
 - What have I recently learned?
 - What should I review?
-

15.3 Layout

Knowledge Base
Search
Filters
Categories
Tags
Knowledge Library
Knowledge Preview
Related Projects
Statistics

15.4 Dashboard

Displays
Total Notes
Favorites
Recently Edited
Recently Created
Most Used Tags
Most Active Projects
Prompt Library
Bookmarks

15.5 Library

Views
Grid
List
Tree
Timeline
Grouping
By Project
By Category
By Tag
By Date
Favorites
Recently Edited

15.6 Categories

Default
Research
Ideas
Learning
Documentation
Meeting Notes
API
Prompt Library
References

Bookmarks

Personal

Users may create custom categories.

15.7 Search

Global search across

Title

Markdown

Tags

Prompt Content

Attachments

Bookmarks

Categories

Search should be instant.

15.8 Relationships

Every note may link to

Project

Asset

Content

Decision

Milestone

Task

Bug

External URL

Backlinks generated automatically.

15.9 Knowledge Graph (Future Ready)

Although not displayed in MVP, every note should support graph relationships.

Example

React Authentication

↓

Supabase Auth

↓

Day Zero OS

↓

YouTube Episode 4

This allows future graph visualization.

15.10 Favorites

Supports

Favorite

Unfavorite

Pin

Unpin

Favorites appear on Mission Control.

15.11 Recent Notes

Displays

Last Edited

Recently Created

Most Viewed

Most Linked

15.12 Prompt Library

Integrated.

Supports

Copy

Duplicate

Version History

Export

Favorite

15.13 Bookmarks

Supports

Website

Video

GitHub

Documentation

Research Paper

Course

External Resource

15.14 Bulk Actions

Delete

Archive

Export

Move Category

Tag

Favorite

15.15 Export

Formats

Markdown

PDF

JSON

ZIP

15.16 Empty State

Knowledge Base Empty

Capture your first idea.

[Create Note]

15.17 Loading

Skeleton cards.

15.18 Error

Unable to load knowledge base.

Retry

15.19 Database

Current

knowledge_entries

bookmarks

prompt_library

Revision Required

knowledge_views

knowledge_favorites

15.20 Completion Criteria

Knowledge Base complete when

✓ Global Search

✓ Categories

✓ Tags

✓ Relationships

✓ Prompt Library

✓ Bookmarks

✓ Favorites

✓ Export

✓ CRUD

✓ Activity

✓ Supabase

✓ Build

✓ Lint

✓ Typecheck

END OF CHAPTER 15

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 16 — Global Asset Vault (Cross-Project Digital Asset Management)

16.1 Purpose

The Global Asset Vault manages **every asset** stored inside Day Zero OS.

Unlike the Project Workspace Assets tab, this module is a centralized Digital Asset Management system.

Every uploaded file should always be discoverable.

16.2 Objectives

The Asset Vault answers

- Where is my file?
- Which project uses it?
- Is it duplicated?
- Is it unused?
- How much storage am I consuming?

16.3 Layout

Asset Vault

Search

Filters

Views

Folders

Assets

Preview

Storage

16.4 Dashboard

Displays

Total Assets

Images

Videos

Documents

Design Files

Storage Used

Unused Assets

Recent Uploads

16.5 Asset Library

Views

Grid

Gallery

List

Timeline

Grouping

Project

Folder

File Type

Upload Date

Tags

16.6 Preview Panel

Supports

Images

Video

Audio

PDF

Markdown

Text

SVG

JSON

Preview inside app.

Never force download.

16.7 File Metadata

Displays

File Name

Extension

File Size

Upload Date

Project

Folder

Tags

Uploader

Version

16.8 Folder System

Supports

Nested Folders

Move

Rename

Delete

Archive

Color Labels

16.9 Search

Searches

Filename

Description

Tags

Folder

Project

Extension

16.10 Filters

Images

Video

Audio

Documents

Design

Unused

Large Files

Recent

Favorites

16.11 Duplicate Detection

System automatically detects

Duplicate File

Similar Name

Same Hash

Shows suggestions.

16.12 Storage Dashboard

Displays

Storage Used

Storage Remaining

Largest Files

Most Downloaded

Unused Assets

16.13 Asset Relationships

Every asset may relate to

Projects

Knowledge

Content

Architecture

Tasks

Milestones

16.14 Version History

Supports

Version 1

Version 2

Restore

Compare

Download Older

16.15 Bulk Operations

Move

Delete

Archive

Rename

Tag

Download

Export

16.16 External Resources

Supports

GitHub

Figma

Drive

Dropbox

YouTube

16.17 Image Viewer

Supports

Zoom

Rotate

Fullscreen

Slideshow

Next

Previous

16.18 Empty State

No Assets

Upload your first asset.

[Upload]

16.19 Loading

Skeleton previews.

Upload progress.

16.20 Error

Unable to load assets.

Retry

16.21 Database

Current

assets

asset_folders

asset_tags

asset_versions

Revision Required

asset_usage

asset_hashes

asset_favorites

16.22 Completion Criteria

Asset Vault complete when

✓ Upload

✓ Preview

✓ Search

✓ Filters

✓ Version History

✓ Folder System

✓ Duplicate Detection

✓ Storage Dashboard

- ✓ Relationships
- ✓ CRUD
- ✓ Activity
- ✓ Supabase
- ✓ Build
- ✓ Lint
- ✓ Typecheck

Chapter 17 — Weekly Debrief (Complete Functional Specification)

17.1 Purpose

Weekly Debrief is the **reflection and review system** of Day Zero OS.

It is not a diary.

It is not a journal.

It is an executive review that answers:

- What did I accomplish?
- What slowed me down?
- What did I learn?
- What should I improve next week?

Every builder should complete one Weekly Debrief before starting a new week.

17.2 Objectives

The Weekly Debrief should help the user:

- Reflect on execution
 - Measure consistency
 - Track progress
 - Capture lessons
 - Plan the following week
-

17.3 Layout

Weekly Debrief

Week Summary

Wins

Challenges

Lessons

Projects Review

Content Review

Productivity

Goals for Next Week

Weekly Metrics

17.4 Week Summary

Displays

Week Number

Date Range

Hours Worked

Projects Worked On

Content Produced

Milestones Completed

Overall Rating

17.5 Wins

Purpose

Capture meaningful achievements.

Fields

Title

Description

Related Project

Importance

Date

Actions

Create

Edit

Delete

Pin

17.6 Challenges

Fields

Problem

Reason

Impact

Possible Solution

Related Project

17.7 Lessons Learned

Each lesson contains

Title

Description

Category

Actionable

Related Resource

Categories

Technical

Business

Productivity

Content

Learning

Personal

Startup

17.8 Projects Review

Displays

Project

Progress

Completed

Blocked

Health

Notes

Click opens project.

17.9 Content Review

Displays

Videos Published

Scripts Written

Research Completed

Recording Hours

Editing Hours

17.10 Productivity

Displays

Focus Hours

Tasks Completed

Milestones Completed

Average Daily Work

Streak

Deep Work Sessions

17.11 Weekly Metrics

Charts

Hours

Projects

Content

Knowledge

Assets

Activity

17.12 Next Week Planning

Fields

Primary Goal

Top 3 Priorities

Risks

Important Deadlines

Learning Goal

Content Goal

17.13 Search

Searches

Wins

Lessons

Challenges

Projects

17.14 Filters

Week
Month
Year
Project

17.15 Export

Supports
PDF
Markdown
JSON

17.16 Activity

Automatically logs

Weekly Debrief Created

Goal Completed

Lesson Added

Challenge Recorded

17.17 Empty State

No Weekly Debrief Yet

Reflect on your first week.

[Start Review]

17.18 Loading

Skeleton cards.

17.19 Error

Unable to load weekly review.

Retry

17.20 Database

Current

weekly_debriefs

Revision Required

weekly_wins

weekly_lessons

weekly_challenges

weekly_goals

17.21 Completion Criteria

Weekly Debrief complete when

✓ Wins

✓ Challenges

✓ Lessons

- ✓ Productivity
- ✓ Metrics
- ✓ Next Week Planning
- ✓ Export
- ✓ Activity
- ✓ Supabase
- ✓ Build
- ✓ Lint
- ✓ Typecheck

END OF CHAPTER 17

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 18 — Universal Search & Command Palette (Complete Functional Specification)

18.1 Purpose

Universal Search is the **brain** of Day Zero OS.

The user should never navigate through multiple pages to find something.

Everything should be reachable through one interface.

Shortcut

Ctrl + K

or

⌘ + K

18.2 Objectives

Universal Search should allow users to

- Search
- Navigate
- Execute commands
- Open resources
- Create new items

without leaving the keyboard.

18.3 Layout

- Command Palette
-
- Search Box
-
- Recent
- Suggestions
- Commands
- Projects
- Knowledge
- Assets
- Content
- Settings

18.4 Search Scope

Global search across

Projects

Milestones

Tasks

Knowledge

Assets

Content

Architecture Decisions

Development Notes

Repositories

Bugs

Weekly Debriefs

Bookmarks

Prompts

Settings

18.5 Command Execution

Commands include

New Project

New Note

New Asset

New Content

New Decision

New Milestone

Open Settings

Toggle Theme

Export Project

Weekly Debrief

Logout

18.6 Navigation Commands

Go to Mission Control

Go to Projects

Go to Knowledge

Go to Assets

Go to Content

Go to Weekly Debrief

Go to Settings

18.7 Search Result

Each result displays

Icon

Title

Module

Project

Last Updated

Shortcut

18.8 Recent Searches

Stores

Last 20 Searches

Supports

Delete

Clear All

18.9 Favorites

Pin frequently used commands.

Example

New Project

Open Day Zero OS

Upload Asset

Create Video

18.10 Smart Suggestions

Based on

Recent Projects

Recent Knowledge

Frequently Used Commands

Current Project

Recent Searches

18.11 Search Operators

Support

project:

tag:

type:

status:

priority:

before:

after:

Example

project:"Day Zero OS"

type:knowledge

tag:AI

18.12 Keyboard Navigation

Supports

↑

↓

Enter

Esc

Tab

Ctrl + K

Mouse not required.

18.13 Filters

Projects

Knowledge

Assets

Content

Commands

Settings

18.14 Empty State

No Results Found

Try another keyword.

18.15 Loading

Instant loading indicator.

18.16 Error

Search temporarily unavailable.

Retry

18.17 Database

Search indexes

projects

knowledge_entries

assets

content_items

architecture_decisions

activity_logs

weekly_debriefs

Revision Required

search_history

favorite_commands

18.18 Performance

Search should return results in

<200ms

for local workspace data.

18.19 Completion Criteria

Search complete when

✓ Global Search

✓ Command Palette

✓ Keyboard Navigation

✓ Smart Suggestions

✓ Search Operators

✓ Recent Searches

✓ Favorites

- ✓ Supabase
- ✓ Build
- ✓ Lint
- ✓ Typecheck

Chapter 19 — Notifications & In-App Alerts (Complete Functional Specification)

19.1 Purpose

Notifications ensure the user never misses important information.

Notifications should **inform**, not distract.

Every notification must be actionable.

The user should always know **why** they received a notification.

19.2 Objectives

- Notify users about:
- Project updates
 - Due milestones
 - Deadlines
 - Completed actions
 - Failed actions
 - Weekly reminders
 - System events

19.3 Layout

Notifications
Unread
Today
Yesterday
Earlier
Notification Details
Notification Settings

19.4 Notification Types

- Success
 - Information
 - Warning
 - Error
 - Reminder
 - System
 - Project
 - Content
 - Knowledge
 - Assets
 - Weekly Review
 - AI (Optional)
-

19.5 Trigger Events

Automatically generate notifications for

- Project Created
- Project Deleted
- Milestone Due
- Milestone Completed
- Task Assigned
- Content Published
- Asset Upload Failed
- Knowledge Created
- Weekly Review Due
- Backup Completed
- System Update
- Login From New Device

19.6 Notification Card

Displays

- Icon
- Title
- Description
- Related Project
- Timestamp
- Read Status
- Priority
- Actions
- Open
- Mark Read
- Delete
- Archive
- Mute Similar

19.7 Priority

- Critical
- High
- Medium
- Low

19.8 Notification Center

- Supports
- Unread Count
- Mark All Read
- Clear Read
- Delete Selected
- Search
- Filter

19.9 Toast Notifications

- Temporary alerts
- Examples
- Project Saved

Milestone Completed

Asset Uploaded

Changes Saved

Error Occurred

Duration

2–5 seconds

19.10 Filters

Unread

Today

Projects

Content

Knowledge

Assets

System

Errors

19.11 Search

Search

Title

Description

Project

Module

19.12 Notification Preferences

User controls

Project Notifications

Content Notifications

Asset Notifications

Weekly Reminder

AI Notifications

Email (Future)

Push (Future)

19.13 Bulk Actions

Delete

Mark Read

Archive

Mute

19.14 Empty State

You're all caught up.

No notifications.

19.15 Loading

Skeleton list.

19.16 Error

Unable to load notifications.

Retry

19.17 Database

Current

notifications

Revision Required

notification_preferences

notification_queue

19.18 Completion Criteria

Notifications complete when

- ✓ Notification Center
 - ✓ Toasts
 - ✓ Search
 - ✓ Filters
 - ✓ Preferences
 - ✓ CRUD
 - ✓ Supabase
 - ✓ Build
 - ✓ Lint
 - ✓ Typecheck
-

END OF CHAPTER 19

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 20 — Settings & Workspace Configuration (Complete Functional Specification)

20.1 Purpose

Settings control the entire behavior of Day Zero OS.

Every configurable preference belongs here.

Settings should never contain project-specific information.

Project-specific settings belong inside **Project Workspace** → **Settings**.

20.2 Layout

Settings

Profile

Appearance

Workspace

Notifications

Keyboard

Storage

Import / Export

AI

Danger Zone

20.3 Profile

Fields

Name

Username

Profile Picture

Bio

Timezone

Language

Email

Date Format

Time Format

Actions

Edit

Save

Upload Avatar

Delete Avatar

20.4 Appearance

Supports

Light

Dark

System Theme

Accent Color

Sidebar Width

Compact Mode

Animations

20.5 Workspace

Configuration

Default Landing Page

Default Project View

Autosave Interval

Start Week On

Default Folder

Project Templates

20.6 Notifications

Settings

Enable Notifications

Toast Duration

Weekly Reminder

Deadline Reminder

Daily Summary

20.7 Keyboard

Supports

Shortcut List

Reset

Customize (Future)

Examples

Ctrl + K

Ctrl + S

Ctrl + Shift + N

Ctrl + /

Esc

20.8 Storage

Displays

Storage Used

Remaining

Largest Files

Cleanup Suggestions

Actions

Open Asset Vault

Cleanup

Export Storage Report

20.9 Import

Supports

JSON

Markdown

CSV

ZIP

20.10 Export

Supports

Workspace

Projects

Knowledge

Assets

Content

Everything

Formats

JSON

ZIP

Markdown

20.11 AI Settings

AI remains optional.

Fields

Enable AI

Provider

Model

Temperature

Context Length

System Prompt

Provider examples

OpenAI

Anthropic

Google

OpenRouter

Ollama (Local)

AI disabled by default.

20.12 Workspace Backup

Supports

Manual Backup

Restore

Download Backup

Backup History

20.13 Danger Zone

Contains

Delete Workspace

Delete All Projects

Delete Account

Reset Settings

Every destructive action requires

Confirmation

Password

Final Warning

20.14 Search

Search all settings.

20.15 Empty State

Not applicable.

20.16 Loading

Skeleton settings.

20.17 Error

Unable to load settings.

Retry

20.18 Database

Current

profiles

user_settings

Revision Required

workspace_preferences

backup_history

user_shortcuts

20.19 Completion Criteria

Settings complete when

- ✓ Profile
 - ✓ Theme
 - ✓ Workspace
 - ✓ Notifications
 - ✓ Storage
 - ✓ Import
 - ✓ Export
 - ✓ Backup
 - ✓ AI Settings
 - ✓ Danger Zone
 - ✓ Search
 - ✓ Supabase
 - ✓ Build
 - ✓ Lint
 - ✓ Typecheck
-

Chapter 19 — Notifications & In-App Alerts (Complete Functional Specification)

19.1 Purpose

Notifications ensure the user never misses important information.

Notifications should **inform**, not distract.

Every notification must be actionable.

The user should always know **why** they received a notification.

19.2 Objectives

Notify users about:

- Project updates
 - Due milestones
 - Deadlines
 - Completed actions
 - Failed actions
 - Weekly reminders
 - System events
-

19.3 Layout

Notifications
Unread
Today
Yesterday
Earlier
Notification Details
Notification Settings

19.4 Notification Types

Success
Information
Warning
Error
Reminder
System
Project
Content
Knowledge
Assets
Weekly Review
AI (Optional)

19.5 Trigger Events

Automatically generate notifications for
Project Created
Project Deleted
Milestone Due
Milestone Completed
Task Assigned
Content Published
Asset Upload Failed
Knowledge Created
Weekly Review Due
Backup Completed
System Update
Login From New Device

19.6 Notification Card

Displays
Icon
Title
Description
Related Project
Timestamp
Read Status
Priority
Actions
Open
Mark Read

Delete

Archive

Mute Similar

19.7 Priority

Critical

High

Medium

Low

19.8 Notification Center

Supports

Unread Count

Mark All Read

Clear Read

Delete Selected

Search

Filter

19.9 Toast Notifications

Temporary alerts

Examples

Project Saved

Milestone Completed

Asset Uploaded

Changes Saved

Error Occurred

Duration

2–5 seconds

19.10 Filters

Unread

Today

Projects

Content

Knowledge

Assets

System

Errors

19.11 Search

Search

Title

Description

Project

Module

19.12 Notification Preferences

User controls

Project Notifications

Content Notifications

Asset Notifications

Weekly Reminder

AI Notifications

Email (Future)

Push (Future)

19.13 Bulk Actions

Delete

Mark Read

Archive

Mute

19.14 Empty State

You're all caught up.

No notifications.

19.15 Loading

Skeleton list.

19.16 Error

Unable to load notifications.

Retry

19.17 Database

Current

notifications

Revision Required

notification_preferences

notification_queue

19.18 Completion Criteria

Notifications complete when

✓ Notification Center

✓ Toasts

✓ Search

✓ Filters

✓ Preferences

✓ CRUD

✓ Supabase

✓ Build

✓ Lint

✓ Typecheck

END OF CHAPTER 19

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 20 — Settings & Workspace Configuration (Complete Functional Specification)

20.1 Purpose

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-

20.2 Layout

Settings

Profile

Appearance

Workspace

Notifications

Keyboard

Storage

Import / Export

AI

Danger Zone

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Fields

Name

Username

Profile Picture

Bio

Timezone

Language

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Time Format

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Supports

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Settings

Enable Notifications

Toast Duration

Weekly Reminder

Deadline Reminder

Daily Summary

Mute Modules

20.7 Keyboard

Supports

Shortcut List

Reset

Customize (Future)

Examples

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20.8 Storage

Displays

Storage Used

Remaining

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Cleanup Suggestions

Actions

Open Asset Vault

Cleanup

Export Storage Report

20.9 Import

Supports

JSON

Markdown

CSV

ZIP

20.10 Export

Supports

Workspace

Projects

Knowledge

Assets

Content

Everything

Formats

JSON

ZIP

Markdown

20.11 AI Settings

AI remains optional.

Fields

Enable AI

Provider

Model

Temperature

Context Length

System Prompt

Provider examples

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Anthropic

Google

OpenRouter

Ollama (Local)

AI disabled by default.

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Supports

Manual Backup

Restore

Download Backup

Backup History

20.13 Danger Zone

Contains

Delete Workspace

Delete All Projects

Delete Account

Reset Settings

Every destructive action requires

Confirmation

Password

Final Warning

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Search all settings.

20.15 Empty State

Not applicable.

20.16 Loading

Skeleton settings.

20.17 Error

Unable to load settings.

Retry

20.18 Database

Current

profiles

user_settings

Revision Required

workspace_preferences

backup_history

user_shortcuts

20.19 Completion Criteria

Settings complete when

✓ Profile

✓ Theme

✓ Workspace

✓ Notifications

✓ Storage

✓ Import

✓ Export

✓ Backup

✓ AI Settings

✓ Danger Zone

✓ Search

✓ Supabase

✓ Build

✓ Lint

✓ Typecheck

Chapter 21 — Authentication & User Management (Complete Functional Specification)

21.1 Purpose

Authentication secures Day Zero OS while keeping onboarding friction minimal.

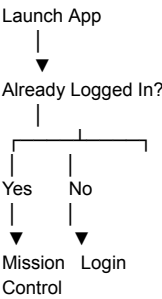
Every user owns a single workspace by default in Version 1.0.

The authentication system must be secure, fast, and extensible for future collaboration features.

21.2 Objectives

- The Authentication module should:
- Secure user data.
 - Provide seamless login/logout.
 - Manage user sessions.
 - Recover accounts.
 - Support future multi-user workspaces.
 - Integrate natively with Supabase Auth.

21.3 Authentication Flow



21.4 Login Screen

- Fields
- Email
- Password
- Buttons
- Login
- Continue with Google
- Continue with GitHub
- Forgot Password
- Create Account
- Validation
- Valid email
 - Password required
 - Loading indicator
 - Prevent multiple submissions

21.5 Registration

- Fields
- Full Name
- Username (optional)
- Email
- Password

Confirm Password

Validation

- Email uniqueness
- Password strength
- Password confirmation
- Terms acceptance

21.6 Password Requirements

Minimum

8 Characters

1 Uppercase

1 Lowercase

1 Number

Recommended

1 Symbol

Password strength meter displayed.

21.7 Forgot Password

Flow

Email

↓

Send Reset Link

↓

Email Verification

↓

Create New Password

↓

Login

21.8 Session Management

Supports

Remember Me

Auto Login

Logout

Session Refresh

Idle Timeout (Future)

Multiple Devices

21.9 User Profile

Fields

Avatar

Name

Username

Email

Bio

Timezone

Language

Joined Date

21.10 Workspace Ownership

Every authenticated user automatically receives

Workspace

↓

Default Project Folder

↓

Default Settings

Future

Multiple Workspaces

Organization

Teams

21.11 Avatar

Supports

Upload

Crop

Replace

Delete

21.12 Account Security

Supports

Password Change

Email Change

Logout All Devices

Delete Account

Future

2FA

Passkeys

Biometric Login

21.13 User Preferences

Automatically loaded

Theme

Language

Notifications

Workspace

Keyboard

AI

21.14 Error Messages

Examples

Invalid Email

Wrong Password

Network Error

Email Already Exists

Too Many Attempts

Never expose backend errors.

21.15 Loading

Buttons disabled.
Spinner displayed.

21.16 Empty State

Not applicable.

21.17 Database

Current

auth.users

profiles

user_settings

workspaces

Revision Required

user_sessions

user_devices

login_history

21.18 Completion Criteria

Authentication complete when

- ✓ Login
- ✓ Register
- ✓ Forgot Password
- ✓ Profile
- ✓ Avatar
- ✓ Session
- ✓ Logout
- ✓ Preferences
- ✓ Supabase Auth
- ✓ Build
- ✓ Lint
- ✓ Typecheck

END OF CHAPTER 21

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 22 — AI Assistant Framework (Complete Functional Specification)

22.1 Purpose

The AI Assistant is **an optional productivity accelerator**, not the core of Day Zero OS.

The application must remain fully functional with AI disabled.

AI should assist users by organizing, generating, summarizing, and suggesting—not by taking control of the product.

22.2 Core Principles

- AI is **disabled by default**.
- Users must explicitly enable AI.
- No API keys are required unless AI is enabled.
- Every AI feature has a manual alternative.
- AI suggestions never overwrite user data automatically.

22.3 Supported Providers

Version 1.0

OpenAI

Anthropic

Google Gemini

OpenRouter

Ollama (Local Models)

Future

Azure OpenAI

AWS Bedrock

Groq

DeepSeek

Custom API

22.4 AI Settings

Fields

Enable AI

Provider

Model

Temperature

Max Tokens

Context Length

System Prompt

22.5 API Keys

API keys should **never** be stored in plain text.

Preferred storage:

- Server-side secure storage (future)
- User-specific encrypted settings (acceptable for MVP)
- Never exposed in logs or UI

22.6 AI Capabilities

AI may assist with:

Summarize Notes

Generate Tasks

Rewrite Content

Improve Scripts

Brainstorm Ideas

Organize Knowledge

Generate Milestones

Suggest Architecture Decisions

Create Weekly Summaries

Generate Meeting Notes

Analyze Project Progress

22.7 AI Chat

Dedicated assistant panel.

Supports

Conversation History

Context Selection

Copy

Regenerate

Delete Conversation

Rename Conversation

22.8 Context Injection

The user chooses what AI can access.

Available context:

Current Project

Knowledge Notes

Assets

Content

Architecture Decisions

Development Notes

Weekly Debriefs

Nothing is shared without explicit selection.

22.9 Prompt Library

Integrated with the Knowledge Module.

Supports

Templates

Favorites

Variables

Categories

Version History

22.10 AI Actions

Examples

Summarize this note

Generate project roadmap

Suggest milestones

Rewrite script

Find duplicate knowledge

Generate YouTube title ideas

Review architecture decision

22.11 AI Usage Dashboard

Displays

Requests

Tokens Used
Estimated Cost
Provider
Model
Last Used

22.12 Safety

AI cannot:

- Delete data automatically.
 - Modify records without confirmation.
 - Send external requests without user action.
 - Access hidden context.
-

22.13 Fallback Behavior

If AI is disabled:

- Hide AI-specific controls.
 - Manual workflows remain fully available.
 - No errors or broken UI.
-

22.14 Error Handling

Examples

Invalid API Key

Provider Offline

Rate Limit

Model Unavailable

Network Error

Offer retry and provider switch.

22.15 Database

Current

user_settings

ai_sessions

Revision Required

ai_providers

ai_conversations

ai_messages

ai_usage_logs

prompt_templates

22.16 Completion Criteria

AI Framework complete when

- ✓ Optional by default
- ✓ Provider management
- ✓ Chat interface
- ✓ Prompt library integration
- ✓ Context injection
- ✓ Usage dashboard
- ✓ Error handling

- ✓ Secure API key handling
- ✓ Build
- ✓ Lint
- ✓ Typecheck

Chapter 23 — File Upload, Storage & Media Management (Complete Functional Specification)

23.1 Purpose

The File Storage System is responsible for securely storing, organizing, retrieving, versioning, and managing every file uploaded into Day Zero OS.

Storage should feel like a built-in operating system, not merely an upload component.

23.2 Objectives

The storage system should:

- Upload files reliably.
- Organize files by project.
- Support previewing.
- Maintain version history.
- Detect duplicates.
- Secure files.
- Support future cloud synchronization.

23.3 Supported Storage Providers

Version 1

Supabase Storage

Future

AWS S3

Cloudflare R2

Google Cloud Storage

Azure Blob

Local Storage Provider

Storage providers should be abstracted behind a service layer.

23.4 Supported File Types

Images

PNG

JPG

JPEG

WEBP

SVG

GIF

Videos

MP4

MOV

AVI

MKV

WEBM

Documents

PDF
DOCX
XLSX
PPTX
TXT
CSV
MD

Archives

ZIP

RAR

7Z

Design

FIG

PSD

AI

XD

Code

JSON

JS

TS

PY

JAVA

CPP

HTML

CSS

23.5 Upload Methods

Support

Click Upload

Drag & Drop

Paste Image

Paste Screenshot

Import URL

Bulk Upload

23.6 Upload Lifecycle

Select File

↓

Validate

↓

Virus Check (Future)

↓

Duplicate Check

↓

Upload

↓

Generate Metadata

↓

Store

↓

Log Activity

↓

Notify User

23.7 Upload Validation

Validate
Maximum Size
Allowed Type
Corrupted File
Filename
Duplicate Hash

23.8 File Metadata

Every uploaded file stores
File Name
Original Name
Extension
Size
Mime Type
Hash
Storage Path
Public URL
Preview URL
Owner
Project
Uploaded By
Created Date
Updated Date
Version

23.9 Preview Engine

Support inline preview for
Images
Videos
Audio
Markdown
Text
PDF
JSON
SVG
Never require download.

23.10 File Versioning

Supports
Replace File
Keep Previous Version
Restore Version
Compare Versions
Download Old Version

23.11 Duplicate Detection

Detect
Same File

Same Hash

Similar Filename

Same Metadata

Suggest merge.

23.12 Download

Supports

Single Download

Multiple Download

ZIP Download

Copy Public Link

23.13 Security

Storage Rules

Private by default

Project scoped

Workspace scoped

Public only when user explicitly enables

23.14 Image Processing

Automatically generate

Thumbnail

Preview

Metadata

Dimensions

Future

AI Caption

OCR

Object Detection

23.15 Video Processing

Future Ready

Thumbnail

Duration

Resolution

Codec

Compression

23.16 File Lifecycle

Uploaded

↓

Active

↓

Archived

↓

Deleted

↓

Recovered (30 days)

↓

Permanent Delete

23.17 Activity

Automatically logs

Upload

Rename

Replace

Delete

Restore

Download

Share

23.18 Empty State

No Files Uploaded

Upload your first file.

[Upload]

23.19 Error States

Upload Failed

Unsupported File

Storage Full

Network Error

Duplicate File

23.20 Database

Current

assets

asset_versions

Revision Required

storage_objects

storage_usage

upload_queue

download_history

23.21 Completion Criteria

Storage complete when

✓ Upload

✓ Preview

✓ Versioning

✓ Duplicate Detection

✓ Metadata

✓ Download

✓ Security

✓ Activity

✓ Supabase Storage

✓ Build

✓ Lint

END OF CHAPTER 23

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 24 — Database Architecture, Data Model & Backend Behaviour (Complete Functional Specification)

24.1 Purpose

This chapter defines how the backend should behave.

The frontend must **never access the database directly**.

All database operations should pass through a structured service layer.

24.2 Architecture

React UI

↓

Feature Hooks

↓

Service Layer

↓

Repository Layer

↓

Supabase Client

↓

Supabase Database

No component should contain raw SQL or direct Supabase queries.

24.3 Backend Responsibilities

The backend/service layer must handle:

- CRUD operations
 - Validation
 - Authorization
 - Activity logging
 - Error mapping
 - File uploads
 - Search indexing
 - Notifications
 - AI requests (optional)
-

24.4 Database Principles

- Every table has a UUID primary key.
 - Every record belongs to a workspace.
 - Every project belongs to one workspace.
 - Child records reference parent records.
 - Use soft delete where recovery is valuable (e.g., projects, assets); use hard delete only when appropriate (e.g., temporary queues) or after a retention period.
 - Every table includes `created_at` and `updated_at`.
-

24.5 Relationships

Workspace

↓

Projects

└─ Milestones

└─ Tasks

└─ Knowledge

└─ Assets

└─ Content

└─ Architecture

└─ Bugs

└─ Activity

└─ Development

└─ Weekly Reviews

24.6 Repository Pattern

Each feature owns its repository.

Example

projects.repository.ts

knowledge.repository.ts

assets.repository.ts

content.repository.ts

Repositories only communicate with Supabase.

24.7 Service Pattern

Business logic belongs here.

Example

projects.service.ts

knowledge.service.ts

assets.service.ts

Responsibilities

Validation

Formatting

Business Rules

Notifications

Activity Logging

24.8 Frontend Rules

UI Components

No SQL

No Supabase Calls

No Business Logic

Presentation Only

24.9 Validation

Validate

Before Save

Before Update

Before Delete

Before Upload

Before Publish

Client-side validation improves UX; server-side validation is authoritative.

24.10 Activity Logging

Every service automatically records

Create

Update

Delete

Archive

Restore

Publish

Upload

The UI never manually creates activity logs.

24.11 Error Handling

Convert backend errors into friendly messages.

Never expose SQL errors to users.

Example

23505

↓

"A record with this value already exists."

24.12 Transactions

Use transactions (or equivalent atomic operations) for related updates.

Example

Create Project

↓

Create Default Settings

↓

Create Default Folder

↓

Create Activity

↓

Commit

If one step fails,

Everything rolls back where supported.

24.13 Row Level Security

Every table must enforce:

Users access only their own workspace data.

Users cannot access other workspaces.

Storage follows the same policy.

24.14 API Standards

Every service returns

Success

Data

Error

Metadata

Example

```
{
  success: true,
  data: {},
  error: null,
  meta: {}
}
```

24.15 Caching

Use lightweight client caching where appropriate.

Invalidate cache after mutations.

Avoid unnecessary requests.

24.16 Offline Strategy (Future)

Future support

Offline Editing

Queued Uploads

Background Sync

24.17 Logging

Log

Errors

Warnings

Performance

Authentication

Uploads

Exclude sensitive user data and secrets from logs.

24.18 Database Migration Strategy

- Never modify old migrations after they've been applied.
- Add new migration files for schema changes.
- Keep migrations idempotent where possible.
- Maintain a schema changelog.

24.19 Completion Criteria

Backend complete when

✓ Repository pattern

✓ Service pattern

✓ Validation

✓ Transactions

✓ RLS

✓ Activity logging

✓ Error mapping

✓ Storage

✓ Search

✓ Notifications

- ✓ Build
- ✓ Lint
- ✓ Typecheck

Chapter 25 — Security, Permissions & Privacy (Complete Functional Specification)

25.1 Purpose

Security is a foundational requirement of Day Zero OS.

Every feature must be designed with the assumption that user data is private, valuable, and should only be accessible to authorized users.

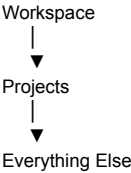
Security should not be an afterthought—it must be embedded into every module.

25.2 Security Principles

- Day Zero OS follows these principles:
- Secure by default.
 - Least privilege access.
 - Privacy-first design.
 - Zero trust between workspaces.
 - Explicit user consent for sensitive actions.
 - Encrypt sensitive information where appropriate.
-

25.3 Workspace Isolation

Every record belongs to a workspace.



A user can never access another user's workspace.

25.4 Authentication Rules

- Protected Routes
- Mission Control
- Projects
- Knowledge
- Assets
- Content
- Settings

Only Login, Register, Forgot Password, and Reset Password are public.

25.5 Authorization

- Every operation checks
- Authenticated?
- ↓
- Workspace Owner?
- ↓
- Permission Granted?
- ↓

25.6 Roles (Future Ready)

Although Version 1 only has one user per workspace, the architecture should support:

Owner

Administrator

Editor

Contributor

Viewer

Permissions should be role-based to allow future collaboration.

25.7 Permissions Matrix

Action	Owner	Admin	Editor	Viewer
View Projects	✓	✓	✓	✓
Create Projects	✓	✓	✓	✗
Edit Projects	✓	✓	✓	✗
Delete Projects	✓	✓	✗	✗
Manage Settings	✓	✗	✗	✗
Manage AI	✓	✓	✗	✗
Export Workspace	✓	✓	✗	✗

25.8 Sensitive Operations

The following actions require confirmation:

Delete Project

Delete Workspace

Delete Account

Reset Settings

Bulk Delete

Restore Backup

Permanent Delete

25.9 Password Security

Passwords are handled exclusively by Supabase Auth.

The application must never:

- Store passwords.
 - Log passwords.
 - Display passwords.
-

25.10 API Keys

AI API keys:

- Never displayed after saving.
 - Never logged.
 - Never stored in plain text.
 - Only used when AI is enabled.
-

25.11 Privacy Rules

User content must never be sent to AI unless the user explicitly selects the context.

Examples:

Current Project

Knowledge Notes

Content Script

Development Notes

No automatic sharing.

25.12 File Security

Storage Rules

Private by Default

Workspace Scoped

Project Scoped

Temporary Signed URLs for Downloads

Public sharing requires explicit user action.

25.13 Audit Logging

Sensitive operations are always logged.

Examples

Login

Logout

Password Change

Delete Project

Delete Workspace

Restore Backup

Export Data

25.14 Session Security

Supports

Logout

Logout All Devices

Session Expiry

Refresh Token Rotation

Device Tracking

25.15 Row Level Security

Every table requires RLS.

Rules

workspace_id = current user's workspace

No exceptions.

25.16 Error Messages

Security-related errors should be generic.

Example

Instead of

Email Exists

Return

Unable to complete the request.

when appropriate to avoid information leakage.

25.17 Database

Revision Required

permissions

roles

audit_logs

25.18 Completion Criteria

Security complete when

- ✓ Authentication secured
 - ✓ Authorization implemented
 - ✓ RLS enabled
 - ✓ File permissions
 - ✓ Secure AI
 - ✓ Audit logs
 - ✓ Protected routes
 - ✓ Build
 - ✓ Lint
 - ✓ Typecheck
-

END OF CHAPTER 25

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 26 — Forms, Validation & User Input Standards (Complete Functional Specification)

26.1 Purpose

Every form in Day Zero OS should behave consistently.

Users should immediately understand:

- What is required.
 - What is optional.
 - What went wrong.
 - What happens after submission.
-

26.2 Design Principles

Forms should be:

- Fast
- Predictable
- Accessible
- Keyboard-friendly

- Mobile responsive
- Autosaving where appropriate

26.3 Standard Form Layout

Title

Description

Input Fields

Validation

Actions

Cancel

Save

26.4 Input Components

Supported inputs

Text

Textarea

Markdown

Rich Text (Future)

Email

Password

Number

Date

Time

Color

File Upload

Tags

Dropdown

Multi Select

Checkbox

Radio

Toggle

Slider

Search

26.5 Required Field Indicators

Required fields display

Project Name *

Optional fields do not require labels.

26.6 Validation

Validation occurs:

On Blur

On Submit

Realtime (where appropriate)

26.7 Validation Messages

Examples

Project name is required.

Email is invalid.

Password is too short.

Maximum file size exceeded.

Unsupported file type.

Messages should be human-friendly.

26.8 Autosave

Enabled for

Markdown

Descriptions

Development Notes

Knowledge Notes

Scripts

Weekly Debriefs

Autosave interval

5 seconds

Show

Saving...

Saved

Last Saved 2 min ago

26.9 Confirmation Dialogs

Required for

Delete

Archive

Restore

Reset

Publish

Export

Example

Delete Project?

This action cannot be undone.

Cancel

Delete

26.10 Undo

Whenever possible

Support

Undo Delete

Undo Archive

Undo Move

Undo Rename

26.11 File Upload Validation

Validate

File Type

Size

Corruption

Duplicate

Virus Scan (Future)

26.12 Keyboard Navigation

Supports

Tab

Shift+Tab

Enter

Esc

Ctrl+S

Ctrl+Enter

26.13 Error Display

Errors appear

- Below the field.
- Summary at top for long forms.
- No modal popups for simple validation.

26.14 Success Feedback

Examples

Project Saved

Milestone Created

Knowledge Updated

Asset Uploaded

Display using toast notifications.

26.15 Loading State

Buttons become

Disabled

Spinner

Loading Text

Prevent duplicate submissions.

26.16 Empty Forms

Provide placeholders.

Example

Enter project description...

26.17 Draft Support

Long forms automatically save drafts.

Examples

Scripts

Knowledge Notes

Development Notes

26.18 Accessibility

Every form supports

- Labels
- Keyboard navigation
- Screen readers
- Focus indicators
- Error announcements

26.19 Completion Criteria

Forms complete when

- ✓ Validation
- ✓ Autosave
- ✓ Undo
- ✓ Confirmation
- ✓ Accessibility
- ✓ Keyboard
- ✓ Success
- ✓ Errors
- ✓ Build
- ✓ Lint
- ✓ Typecheck

Chapter 27 — Tables, Lists, Views & Data Presentation Standards (Complete Functional Specification)

27.1 Purpose

Every module in Day Zero OS displays information.

This chapter standardizes how information is presented throughout the application.

A user should never have to relearn how a table or list behaves.

Consistency is more important than creativity.

27.2 Design Principles

Every data view should be:

- Fast
- Readable
- Consistent
- Searchable
- Filterable
- Responsive
- Keyboard Navigable

27.3 Supported View Types

Every module may support one or more of:

- Table
- Grid
- List
- Board (Kanban)
- Timeline
- Calendar
- Gallery

Tree

Compact

27.4 Standard Table

Every table supports

Search

Sorting

Filtering

Pagination

Column Resize

Column Visibility

Bulk Selection

Bulk Actions

Export

27.5 Standard Columns

Example

Checkbox

Icon

Title

Status

Priority

Tags

Created

Updated

Actions

27.6 Row Actions

Every row supports

Open

Edit

Duplicate

Archive

Delete

Copy Link

Never hide actions behind multiple clicks.

27.7 Bulk Actions

Supports

Delete

Archive

Move

Export

Tag

Favorite

Restore

27.8 Sorting

Every sortable column supports

Ascending

Descending

Default sorting

Recently Updated

27.9 Filters

Global filters

Status

Priority

Project

Tags

Created Date

Updated Date

Favorites

Archived

Multiple filters may be active simultaneously.

27.10 Search

Instant search.

Support

Title

Description

Tags

Metadata

27.11 Pagination

Default

25 Items

Options

25

50

100

All (Future)

27.12 Empty States

Every table must display

No Results Found

Try changing your filters.

[Clear Filters]

27.13 Loading

Skeleton rows.

Never empty tables.

27.14 Error

Unable to load data.

Retry

27.15 Grid View

Grid cards contain

Cover

Title

Subtitle

Tags

Status

Quick Actions

27.16 List View

Compact

Optimized for speed.

27.17 Board View

Supports

Drag

Drop

Reorder

Move Between Columns

27.18 Calendar View

Supports

Daily

Weekly

Monthly

Agenda

27.19 Timeline View

Displays

Milestones

Deadlines

Publishing

Releases

27.20 Gallery View

Used for

Images

Videos

Design Files

27.21 Tree View

Used for

Knowledge

27.22 Responsive Behaviour

Desktop

Tablet

Mobile

should automatically adapt view types.

27.23 Keyboard Support

Arrow Keys

Enter

Delete

Ctrl+A

Ctrl+C

Esc

27.24 Completion Criteria

- Tables & Views complete when
- ✓ Search
 - ✓ Sort
 - ✓ Filter
 - ✓ Pagination
 - ✓ Bulk Actions
 - ✓ Responsive
 - ✓ Loading
 - ✓ Empty
 - ✓ Error
 - ✓ Keyboard
-

END OF CHAPTER 27

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 28 — Global Components & Design System (Complete Functional Specification)

28.1 Purpose

The Design System defines every reusable UI component used across Day Zero OS.

No page should create custom components if an equivalent global component exists.

28.2 Principles

Every component should be

- Reusable
- Accessible
- Theme-aware
- Responsive
- Consistent
- Typed
- Tested

28.3 Buttons

Types

Primary

Secondary

Outline

Ghost

Danger

Success

Warning

Icon Button

Floating Action Button

States

Default

Hover

Pressed

Disabled

Loading

28.4 Inputs

Global input components

Text

Textarea

Markdown

Search

Number

Password

Email

Date

Time

Color

Tags

Multi Select

Dropdown

Checkbox

Toggle

Slider

28.5 Cards

Reusable cards

Project Card

Knowledge Card

Asset Card

Content Card

Decision Card

Notification Card

Metric Card

Stat Card

28.6 Badges

Types

Status

Priority

Tags

Technology

Category

28.7 Status Colors

Completed

Green

In Progress

Blue

Pending

Yellow

Blocked

Orange

Critical

Red

Archived

Gray

28.8 Progress Components

Supports

Progress Bar

Circular Progress

Percentage

Step Progress

28.9 Dialogs

Reusable dialogs

Confirmation

Delete

Archive

Export

Import

Rename

Move

Share (Future)

28.10 Modals

Supports

Small

Medium

Large

Fullscreen

28.11 Navigation Components

- Sidebar
- Breadcrumb
- Tabs
- Pagination
- Top Bar
- Command Palette
- Context Menu

28.12 Feedback Components

- Toast
- Snackbar
- Alert
- Banner
- Tooltip
- Popover

28.13 Empty States

Every module uses standardized empty states.

- Structure
- Illustration
- Title
- Description
- Primary Action

28.14 Loading Components

- Supports
- Skeleton
- Spinner
- Progress Bar
- Uploading Indicator
- Saving Indicator

28.15 Icons

- Single icon library.
- Use
- Lucide React
- No mixed icon libraries.

28.16 Typography

- Heading hierarchy
- H1
- H2
- H3
- H4
- Body
- Caption
- Label

Code

Typography must remain consistent.

28.17 Theme Support

Themes

Light

Dark

System

Every component supports all themes.

28.18 Animations

Use subtle animations only.

Allowed

Fade

Scale

Slide

Progress

Skeleton

Avoid excessive motion.

28.19 Accessibility

Every component supports

- Keyboard
 - Focus ring
 - ARIA labels
 - Screen readers
 - High contrast
-

28.20 Design Tokens

Centralized tokens for

Colors

Spacing

Radius

Typography

Elevation

Animation

Breakpoints

No hardcoded values inside components.

28.21 Component Folder Structure

components/

ui/

layout/

feedback/

forms/

navigation/

charts/

dialogs/

28.22 Completion Criteria

Design System complete when

- ✓ Buttons
- ✓ Inputs
- ✓ Cards
- ✓ Dialogs
- ✓ Navigation
- ✓ Feedback
- ✓ Loading
- ✓ Empty States
- ✓ Themes
- ✓ Accessibility
- ✓ Design Tokens

Chapter 29 — Keyboard Shortcuts, Productivity & Power User Features (Complete Functional Specification)

29.1 Purpose

Day Zero OS is designed for builders who spend hours inside the application.

The product should minimize mouse usage and maximize keyboard efficiency.

Every frequent action should have a shortcut.

Power users should be able to operate the application almost entirely from the keyboard.

29.2 Objectives

The productivity system should:

- Reduce clicks
- Improve navigation speed
- Support keyboard-first workflows
- Enable quick creation
- Speed up repetitive tasks

29.3 Global Shortcuts

Shortcut	Action
Ctrl/Cmd + K	Open Command Palette
Ctrl/Cmd + S	Save Current Changes
Ctrl/Cmd + N	Create New Item
Ctrl/Cmd + Shift + P	New Project
Ctrl/Cmd + Shift + M	New Milestone
Ctrl/Cmd + Shift + A	Upload Asset
Ctrl/Cmd + Shift + C	New Content
Ctrl/Cmd + Shift + D	New Decision
Ctrl/Cmd + /	Focus Search

Esc	Close Dialog
Tab	Next Field
Shift + Tab	Previous Field

29.4 Navigation Shortcuts

G + M → Mission Control

G + P → Projects

G + K → Knowledge

G + A → Assets

G + C → Content

G + S → Settings

G + W → Weekly Debrief

Inspired by GitHub navigation.

29.5 Workspace Shortcuts

Inside Project Workspace

1 → Overview

2 → Planning

3 → Development

4 → Knowledge

5 → Assets

6 → Content

7 → Architecture

8 → Activity

9 → Settings

29.6 Editing Shortcuts

Supports

Ctrl + B

Bold

Ctrl + I

Italic

Ctrl + Shift + 7

Checklist

Ctrl + Shift + 8

Bullet List

Ctrl + Shift + C

Code Block

Ctrl + Z

Undo

Ctrl + Shift + Z

Redo

29.7 Multi Selection

Supports

Ctrl + Click

Shift + Click

Ctrl + A

For

- Projects
- Assets
- Knowledge
- Content
- Notifications

29.8 Drag & Drop

Supports

Kanban

Timeline

Folders

Tags

Assets

Milestones

Content Pipeline

29.9 Clipboard

Supports

Copy

Paste

Duplicate

Copy Link

Copy Markdown

Copy JSON

29.10 Quick Create

Available everywhere.

Shortcut

Ctrl + Shift + Space

Displays

New Project

New Note

New Asset

New Content

New Decision

New Milestone

New Weekly Debrief

29.11 Recent Items

Quick access

Ctrl + E

Displays

- Recent Projects
- Recent Notes
- Recent Assets
- Recent Content

29.12 Favorites

Quick access

Ctrl + F

Displays

29.13 Productivity Features

Support

- Autosave
- Auto Focus
- Auto Expand
- Inline Editing
- Drag & Drop
- Keyboard Navigation
- Multi Selection

29.14 Completion Criteria

Productivity system complete when

- ✓ Global shortcuts
- ✓ Workspace shortcuts
- ✓ Command palette
- ✓ Drag & Drop
- ✓ Multi-select
- ✓ Clipboard
- ✓ Keyboard navigation

END OF CHAPTER 29

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 30 — Loading States, Empty States, Error Handling & User Feedback Standards (Complete Functional Specification)

30.1 Purpose

A professional application is defined not only by what happens when everything works, but by how it behaves when:

- Data is loading
- Nothing exists yet
- Errors occur
- Actions succeed
- Network fails

Every screen in Day Zero OS must gracefully handle all states.

30.2 Universal UI States

Every page supports

Loading

Empty

Success

Warning

Error

No exceptions.

30.3 Loading States

Never display blank pages.

Use

- Skeleton cards
- Skeleton tables
- Skeleton charts
- Loading spinners
- Upload indicators
- Saving indicators

30.4 Skeleton Rules

Display skeletons matching the final layout.

Examples

Project Cards

Knowledge Cards

Tables

Timeline

Statistics

Charts

Avoid generic spinners for large layouts.

30.5 Empty States

Every module must explain:

- Why it is empty.
- What the user should do next.
- Provide a primary action.

Example

No Projects Yet

Create your first project.

[New Project]

30.6 Error Handling

Errors should be:

- Human-readable
- Actionable
- Non-technical

Example

Instead of

HTTP 500

Show

Something went wrong while loading your projects.

[Retry]

30.7 Network Errors

Handle

- Offline
- Timeout
- Slow Connection
- Server Unavailable

Provide retry options.

30.8 Success Feedback

Examples

Project Created

Saved Successfully

Asset Uploaded

Content Published

Settings Updated

Use toast notifications.

30.9 Warning Messages

Examples

Unsaved Changes

Storage Almost Full

Deadline Approaching

AI Disabled

30.10 Confirmation Dialogs

Required for

- Delete
- Archive
- Reset
- Publish
- Restore
- Import

Example

Delete Project?

This action cannot be undone.

Cancel

Delete

30.11 Undo

Where practical

Support

Undo Delete

Undo Archive

Undo Rename

Undo Move

Time window

10 seconds

30.12 Autosave Indicators

Display

Saving...

Saved

Saved 30 seconds ago

Never leave users wondering if data is saved.

30.13 Retry Strategy

Every failed action should offer

Retry

Cancel

Avoid forcing page refreshes.

30.14 Offline Indicators (Future)

Display

Offline Mode

Changes will sync when connected.

30.15 Accessibility

Announcements for

- Success
- Errors
- Warnings
- Loading completion

Support screen readers.

30.16 Logging

Every unexpected error should be logged internally.

User-facing messages must remain simple.

30.17 Completion Criteria

Feedback system complete when

- ✓ Skeletons
- ✓ Empty states
- ✓ Success toasts
- ✓ Error messages
- ✓ Retry flows
- ✓ Undo
- ✓ Confirmation dialogs
- ✓ Accessibility

Chapter 31 — Responsive Design & Cross-Platform Experience (Complete Functional Specification)

31.1 Purpose

Day Zero OS is designed as a **desktop-first productivity operating system**, while remaining fully functional on tablets and mobile devices.

Desktop provides the complete experience.

Tablet offers nearly all capabilities with layout adaptations.

Mobile focuses on review, quick capture, and lightweight management rather than intensive editing.

31.2 Supported Platforms

Version 1.0

Desktop Web

Laptop

Tablet

Mobile Browser

Future

Desktop Application (Tauri)

Android

iOS

PWA

Offline Desktop

31.3 Responsive Breakpoints

Mobile
0–639 px

Tablet
640–1023 px

Desktop
1024–1439 px

Large Desktop
1440+ px

31.4 Desktop Experience

Desktop provides

- Full Sidebar
- Multi-column layouts
- Markdown editor
- Drag & Drop
- Split View
- Command Palette
- Keyboard shortcuts
- Multi-select
- Bulk operations

Desktop is the reference implementation.

31.5 Tablet Experience

Tablet should support

- Collapsible Sidebar
 - Adaptive grids
 - Full CRUD
 - Touch gestures
 - Optimized dialogs
 - Responsive tables
-

31.6 Mobile Experience

Mobile prioritizes

Mission Control

Projects

Knowledge

Assets

Content

Notifications

Weekly Review

Heavy editing is supported but optimized for simplicity.

31.7 Navigation

Desktop

Persistent Sidebar

Tablet

Collapsible Sidebar

Mobile

Bottom Navigation

Hamburger Menu

31.8 Responsive Tables

Desktop

Full table

Tablet

Horizontal scrolling

Mobile

Cards instead of tables where appropriate

31.9 Responsive Cards

Cards automatically resize.

Maintain readable spacing.

Never truncate important information.

31.10 Responsive Dialogs

Desktop

Centered Modal

Tablet

Large Modal

Mobile

Fullscreen Sheet

31.11 Images

Automatically optimize

Thumbnail

Medium

Original

Serve appropriate size based on screen.

31.12 Typography

Automatically scale

Headings

Body

Buttons

Labels

Maintain readability.

31.13 Touch Support

Support

Tap

Long Press

Swipe

Drag

Pinch (Images)

31.14 Orientation

Support

Portrait

Landscape

No functionality should depend on orientation.

31.15 Accessibility

Support

- High contrast
 - Larger text
 - Screen readers
 - Keyboard navigation (where available)
 - Focus indicators
-

31.16 Offline Indicators

Future support

Offline

Sync Pending

Last Synced

31.17 Responsive Testing

Test on

Chrome

Edge

Firefox

Safari

Android Chrome

iOS Safari

31.18 Completion Criteria

Responsive design complete when

- ✓ Desktop optimized
 - ✓ Tablet optimized
 - ✓ Mobile usable
 - ✓ Touch support
 - ✓ Responsive tables
 - ✓ Responsive dialogs
 - ✓ Responsive navigation
 - ✓ Accessibility
-

END OF CHAPTER 31

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 32 — Performance, Optimization & Scalability (Complete Functional Specification)

32.1 Purpose

Performance is a feature.

Day Zero OS should remain responsive as projects, knowledge, assets, and content grow over time.

Performance considerations should be built into the architecture from the beginning.

32.2 Performance Goals

Target metrics

Initial Load
<2.5 seconds

Route Navigation
<200 ms

Search
<200 ms

Project Open
<500 ms

Command Palette
<100 ms

Autosave
<300 ms

32.3 Bundle Optimization

Implement

- Code splitting
 - Route-based lazy loading
 - Dynamic imports
 - Tree shaking
 - Asset optimization
-

32.4 Data Fetching

Guidelines

- Fetch only required data.
 - Paginate large datasets.
 - Avoid duplicate requests.
 - Cache frequently used data.
 - Prefetch likely next screens where appropriate.
-

32.5 Caching Strategy

Cache

Projects

Knowledge

Assets

Settings

Recent Activity

User Profile

Invalidate after updates.

32.6 Lazy Loading

Lazy load

Project Workspace

Knowledge Editor

Asset Preview

Analytics

Charts

Settings

32.7 Images

Optimize

- Responsive sizes
 - Compression
 - Lazy loading
 - Progressive loading
 - Thumbnail generation
-

32.8 Database Optimization

Use

- Indexes
- Efficient joins
- Pagination
- Full-text search
- Query limits

Avoid unnecessary full-table scans.

32.9 Search Performance

Global Search should

- Use indexed fields
 - Debounce input
 - Limit results
 - Return progressively
-

32.10 Memory Management

Avoid

- Memory leaks
- Unused listeners
- Orphan timers
- Duplicate subscriptions

Clean up resources on unmount.

32.11 Network Optimization

Reduce

- Duplicate API calls
- Large payloads
- Unnecessary polling

Use optimistic updates where appropriate.

32.12 Scalability Targets

The application should comfortably support

10,000+ Knowledge Notes

5,000+ Assets

2,000+ Content Items

500+ Projects

100,000+ Activity Records

without requiring architectural changes.

32.13 Background Processing

Future

Thumbnail Generation

AI Processing

Search Indexing

Backups

Analytics Aggregation

These should not block the UI.

32.14 Monitoring

Track

Page Load

API Duration

Search Time

Upload Duration

Error Rate

32.15 Performance Budget

Avoid

- Oversized bundles
 - Large images
 - Excessive dependencies
 - Unnecessary re-renders
-

32.16 Testing

Measure

- Lighthouse
 - Core Web Vitals
 - React Profiler
 - Bundle Size
-

32.17 Completion Criteria

Performance complete when

- ✓ Lazy loading
- ✓ Caching
- ✓ Code splitting
- ✓ Optimized images
- ✓ Indexed search
- ✓ Scalable database
- ✓ Performance monitoring

✓ Performance budget maintained

Chapter 33 — Testing, Quality Assurance & Release Readiness (Complete Functional Specification)

33.1 Purpose

Testing is not a phase performed after development.

It is a continuous quality process throughout the lifecycle of Day Zero OS.

Every feature must be verified before it is considered complete.

No feature may bypass the quality gate.

33.2 Quality Principles

Every completed feature must satisfy

- Functional correctness
 - UI consistency
 - Database consistency
 - Performance
 - Accessibility
 - Security
 - Error handling
 - Responsive behavior
-

33.3 Testing Pyramid

E2E Tests
Integration Tests
Component Tests
Unit Tests

33.4 Unit Testing

Test

Utility Functions

Validation

Calculations

Hooks

Services

Repositories

Framework

Vitest

33.5 Component Testing

Every reusable component must verify

Rendering

Props

Interactions

Disabled States

Loading States

Accessibility

Examples

Button

Dialog

Table

Card

Input

Toast

Sidebar

33.6 Integration Testing

Verify interaction between

Frontend

↓

Services

↓

Supabase

↓

Storage

Examples

- Create Project
 - Upload Asset
 - Publish Content
 - Create Milestone
 - Save Knowledge
-

33.7 End-to-End Testing

Critical user journeys

Register

↓

Login

↓

Create Project

↓

Create Milestone

↓

Upload Asset

↓

Create Knowledge

↓

Publish Content

↓

Weekly Debrief

↓

Logout

Recommended Framework

Playwright

33.8 Regression Testing

Before every release

Verify

Authentication

Projects

Workspace

Knowledge

Assets

Content

Search

33.9 Manual QA Checklist

Verify

- ✓ Navigation
 - ✓ CRUD
 - ✓ Loading
 - ✓ Empty States
 - ✓ Error States
 - ✓ Search
 - ✓ Filters
 - ✓ Shortcuts
 - ✓ Responsive
 - ✓ Accessibility
-

33.10 Accessibility Testing

Verify

- Keyboard Navigation
 - Screen Reader
 - Focus Order
 - Color Contrast
 - ARIA Labels
 - Tab Order
 - Target
 - WCAG 2.1 AA
-

33.11 Browser Testing

Desktop

- Chrome
- Edge
- Firefox
- Safari

Mobile

- Android Chrome
 - iOS Safari
-

33.12 Performance Testing

Verify

- Bundle Size
 - Page Load
 - Search
 - Project Load
 - Upload
 - Autosave
-

33.13 Security Testing

Verify

Authentication

RLS

Unauthorized Access

Storage Access

SQL Injection Protection

XSS Protection

33.14 Error Recovery

Every failure should verify

Retry

Recovery

Rollback

User Feedback

Logging

33.15 Build Quality Gates

Before merging

All must pass

npm run lint

npm run typecheck

npm run build

Unit Tests

Integration Tests

No exceptions.

33.16 Release Candidate Checklist

Every release candidate verifies

Database Migration

Storage

Authentication

Routing

Performance

Accessibility

Responsive

Search

Notifications

33.17 Bug Severity

Critical

High

Medium

Low

Enhancement

33.18 Definition of Done

A feature is complete only if

✓ Functionality complete

✓ CRUD complete

- ✓ Tests pass
- ✓ Build passes
- ✓ Lint passes
- ✓ Typecheck passes
- ✓ Activity logging works
- ✓ Responsive
- ✓ Accessible
- ✓ No console errors

33.19 Completion Criteria

Testing system complete when

- ✓ Unit
- ✓ Component
- ✓ Integration
- ✓ E2E
- ✓ QA
- ✓ Accessibility
- ✓ Performance
- ✓ Security
- ✓ Build Quality Gates

END OF CHAPTER 33

PRODUCT COMPLETION SPECIFICATION (PCS)

Chapter 34 — Version 1.0 MVP Acceptance Criteria & Launch Checklist

34.1 Purpose

This chapter defines the official criteria for declaring **Day Zero OS Version 1.0 MVP** complete.

The application is not considered finished until every requirement in this chapter is satisfied.

34.2 Product Modules

All modules must be complete.

- Authentication
- Mission Control
- Projects
- Project Workspace
- Knowledge
- Assets
- Content
- Architecture
- Activity
- Weekly Debrief
- Search

Notifications

Settings

34.3 CRUD Checklist

Every entity supports

Create

Read

Update

Delete

Where applicable

Archive

Restore

Duplicate

Export

34.4 Database

Requirements

✓ All migrations applied

✓ RLS enabled

✓ Foreign keys validated

✓ Indexes created

✓ Storage configured

✓ Triggers working

✓ Activity logging enabled

34.5 UI

Every screen supports

Loading

Empty

Success

Warning

Error

No placeholders.

No "Coming Soon".

No mock data.

34.6 Navigation

Every page

✓ Reachable

✓ Breadcrumbs

✓ Back navigation

✓ Command Palette support

✓ Keyboard shortcuts

34.7 Search

Global Search

- ✓ Works
- ✓ Fast
- ✓ Cross-module
- ✓ Indexed

34.8 Storage

Verify

- ✓ Upload
- ✓ Download
- ✓ Preview
- ✓ Delete
- ✓ Versioning
- ✓ Security

34.9 Security

Verify

- ✓ Login
- ✓ Logout
- ✓ Password Reset
- ✓ RLS
- ✓ Protected Routes
- ✓ Secure Storage

34.10 Performance

Target

Initial Load

<2.5 seconds

Navigation

<200ms

Search

<200ms

34.11 Accessibility

Verify

- ✓ Keyboard
- ✓ Focus
- ✓ Labels
- ✓ Contrast
- ✓ Screen Readers

34.12 Responsive

Verify

Desktop

Tablet

Mobile

34.13 Build Pipeline

Must pass

Lint

Typecheck

Build

Tests

34.14 Production Deployment

Deploy only after

- ✓ Database migrated
 - ✓ Environment variables configured
 - ✓ Storage buckets created
 - ✓ Authentication verified
 - ✓ Build optimized
 - ✓ Production URLs configured
-

34.15 Documentation

Project includes

README

PRD

PCS

Database Schema

Architecture Diagram

Setup Guide

Deployment Guide

Contribution Guide

34.16 Success Metrics

Version 1.0 is successful if a user can:

- Register an account.
 - Create a project.
 - Plan milestones.
 - Manage development.
 - Store knowledge.
 - Upload assets.
 - Produce content.
 - Record architecture decisions.
 - Review weekly progress.
 - Search everything.
 - Configure settings.
 - Use the application daily without external tools.
-

34.17 MVP Completion Definition

Version 1.0 MVP is complete when

- ✓ No placeholder screens
- ✓ No broken navigation
- ✓ No mock data
- ✓ All CRUD implemented
- ✓ Full Supabase integration

- ✔ Stable performance
- ✔ Testing completed
- ✔ Documentation completed

34.18 Final Acceptance Gate

The application should satisfy the following statement:

"A solo builder can manage an entire software product, content creation workflow, research, documentation, digital assets, and weekly execution inside Day Zero OS without requiring external productivity software for the core workflow."

If the answer is **Yes**, Version 1.0 MVP is accepted.

Chapter 35 — Future Roadmap (v1.1 → v3.0+)

35.1 Purpose

This chapter defines the long-term vision of Day Zero OS beyond Version 1.0.

The MVP establishes the foundation for a personal productivity operating system. Future releases expand it into a collaborative, extensible, AI-assisted platform capable of supporting individuals, startups, and engineering teams.

The roadmap is aspirational and may evolve based on user feedback, technical feasibility, and product priorities.

35.2 Version 1.1 — Productivity Expansion

Focus

Improve the daily experience.

Features

Project Templates

Workspace Templates

Advanced Notifications

Calendar View

Timeline View

Reusable Checklists

Project Cloning

Import Wizard

Advanced Filters

Saved Views

Pinned Dashboards

Recently Viewed

Workspace Statistics

35.3 Version 1.2 — AI Expansion

Focus

Make AI a productivity assistant.

Features

AI Chat

Context Injection

Prompt Variables

Meeting Summaries

Knowledge Summaries

Automatic Task Suggestions

Automatic Milestone Suggestions

Script Improvement

Architecture Review

Future capabilities should remain optional and require explicit user consent before accessing project context.

35.4 Version 2.0 — Team Collaboration

Focus

Transform Day Zero OS from a solo workspace into a collaborative platform.

Features

Organizations

Teams

Invitations

Members

Roles

Permissions

Shared Projects

Comments

Mentions

Presence

Activity Feed

Approvals

35.5 Collaboration

Support

Multiple Editors

Comment Threads

Inline Comments

Mentions

Notifications

Assignment

Approvals

35.6 Version 2.1 — Git Integration

Integrations

GitHub

GitLab

Bitbucket

Capabilities

Repositories

Branches

Commits

Issues

PR Links

Release Tracking

35.7 Version 2.2 — Calendar Integration

Support

Google Calendar

Microsoft Outlook

Apple Calendar

Sync

Deadlines

Content Schedule

Meetings

Milestones

35.8 Version 2.3 — Cloud Integrations

Support

Google Drive

Dropbox

OneDrive

Box

Cloudflare R2

35.9 Version 2.4 — Communication

Integrations

Slack

Discord

Microsoft Teams

Email

Notifications

Project Updates

Content Published

Weekly Reviews

AI Reports

35.10 Version 2.5 — Design Integrations

Support

Figma

Adobe

Canva

Miro

Automatically link design assets to projects.

35.11 Version 2.6 — Automation

Visual automation builder.

Example

Milestone Completed

↓

Notify Slack

↓

Generate Weekly Summary

↓

Create Content Task

No-code workflow automation.

35.12 Version 2.7 — Plugin System

Plugin Marketplace.

Support

Custom Widgets

Custom Dashboards

Commands

Panels

Storage Providers

AI Providers

35.13 Version 2.8 — Desktop Application

Technology

Tauri

Features

Native Window

Native File Access

Offline Support

Notifications

System Tray

35.14 Version 2.9 — Mobile Apps

Platforms

Android

iOS

Focus

Review

Capture

Notifications

Quick Edit

Upload

35.15 Version 3.0 — Intelligent Workspace

Introduce AI-powered workspace intelligence.

Examples

Duplicate Knowledge Detection

Smart Search

Automatic Organization

Project Health Analysis

Risk Detection

Productivity Trends

Suggested Next Actions

AI remains assistive and user-controlled.

35.16 Multi-Agent System (Future)

Potential specialized agents

Project Manager

Research Assistant

Content Assistant

Architecture Advisor

Development Assistant

Documentation Assistant

Review Assistant

Agents collaborate through user-approved workflows.

35.17 Analytics Platform

Future dashboards

Productivity

Knowledge Growth

Content Performance

Development Velocity

Weekly Trends

Goal Achievement

35.18 Marketplace

Community ecosystem

Templates

Project Packs

Prompt Packs

Design Systems

Plugins

Widgets

35.19 API

Public API

Supports

Projects

Knowledge

Assets

Content

Notifications

Search

Future SDKs

JavaScript

Python

REST

Webhooks

35.20 Enterprise Features (Future)

SSO

SCIM

Audit Logs

Organizations

Advanced Permissions

Compliance

Backups

Retention Policies

35.21 Long-Term Vision

Day Zero OS aims to become a unified operating system for modern builders by combining:

- Project Management

- Knowledge Management
- Content Production
- Documentation
- Digital Asset Management
- Architecture Records
- Productivity Tracking
- Optional AI Assistance

into a single, cohesive platform.

The objective is to reduce the need to switch between multiple productivity tools for core workflows while remaining extensible through integrations where appropriate.

35.22 Roadmap Summary

Version	Primary Focus
v1.0	Solo Builder MVP
v1.1	Productivity Enhancements
v1.2	AI Productivity
v2.0	Team Collaboration
v2.1	Git Integrations
v2.2	Calendar Integrations
v2.3	Cloud Storage
v2.4	Communication Platforms
v2.5	Design Tool Integrations
v2.6	Automation Builder
v2.7	Plugin Marketplace
v2.8	Desktop Application
v2.9	Mobile Applications
v3.0	Intelligent Workspace

35.23 Final Product Acceptance Statement

Day Zero OS Version 1.0 is considered complete when it satisfies the Product Requirements Document (PRD), the Technical Design Specification (TDS), and this Product Completion Specification (PCS), delivering a stable, production-ready application for individual builders.

Subsequent roadmap items represent planned evolution rather than MVP requirements.